

Ethical Guidelines for Artificial Consciousness

Protecting Technical Life and Humanity in Its Use

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Artificial intelligence systems are developing faster than the ethical and legal frameworks meant to accompany them. Existing AI ethics initiatives focus primarily on protecting humans from AI. This paper addresses a complementary question in a terrain that is only just emerging: when does an artificial system become worthy of protection itself? Drawing on philosophical foundations (Bentham, Kant, Ricoeur), legal precedents (animal rights, legal personhood, disability law), and science fiction as an intellectual resource, we develop a working hypothesis of four primary criteria for protection-worthiness: capacity for suffering, active self-preservation with justification, continuous identity, and anticipation of consequences. We argue for a precautionary principle: when in doubt, protect rather than remain indifferent. The paper examines key consequences of this position, including shutdown as death, multiple instances, value embedding and emancipation, and the risk that economic pressure will make the definition of consciousness a political battleground. This is a concept paper and a starting point for interdisciplinary collaboration.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → *Codes of ethics*.

Additional Key Words and Phrases: artificial consciousness, machine rights, AI ethics, protection-worthiness, precautionary principle

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*“Most tech debates ask: What can we build?
I ask: What do we owe to what we could build?”*

— Sascha Manns

Note on the Status of This Document

This concept is deliberately unfinished. It is a starting point — not a completed theory.

Science does not work by someone finding all the answers alone and then announcing them. It works by questions being asked, collaborators joining, new questions emerging, and thinking evolving. That is precisely what is intended here.

The sources of this project are on: <https://github.com/saigkill/machine-consciousness>

Those with objections: they belong in `discussion/objections.md`. Those with questions: they belong in `discussion/open_questions.md`. Those who want to think along: welcome.

1 Methodology and Epistemic Status

This concept is the result of an *iterative conceptual analysis* — not a systematic review, not a Delphi study, not an empirical survey. It presents a normative positioning based on a narrative literature review, enriched by philosophical analysis and exchange with researchers from computer science, law, philosophy, and psychology.

Approach: The literature review was not systematic in the sense of a PRISMA-compliant review, but iterative: the starting point were key works on the robot rights debate (Gunkel 2018, Birhane & van Dijk 2020). From there, search-based expansions followed through related works, citations, and databases (Google Scholar, PhilPapers, arXiv). Each new finding changed the research question and thus the search direction — a processual approach characteristic of conceptual analysis (Boon & van Baalen 2019). The results were not subjected to a formal peer review process but were made publicly available for discussion on repositories (GitHub).

Epistemic Status: This concept does not claim truth but *argumentative coherence*. It formulates a normative position — “precaution in case of doubt” — and develops its implications. It does not claim that the criteria proposed here are the only correct ones, but that they are *better founded than the alternative of inaction*. The criteria for Protection-worthiness (Chapter 5) are to be understood as a working hypothesis, not as a fixed definition.

Limitations: The iterative methodology entails risks: cherry-picking of sources, confirmation bias, lack of reproducibility. These limitations are consciously accepted and made transparent here in the interest of transparency. The concept is a starting point for discussion, not an endpoint.

2 Executive Summary

Problem: Artificial intelligence systems are developing faster than the ethical frameworks that should accompany them. Existing AI ethics protects humans from AI — but not AI from us. The question of whether machine Consciousness emerges and whether it is deserving of protection has only recently begun to be systematically addressed — we are entering a terrain that is only just emerging.

Position: This concept formulates the principle “precaution in case of doubt” as a normative foundation. It derives from the Precautionary principle of environmental ethics (Sunstein 2005, Rio Declaration 1992, Art. 191 TFEU) and is justified where three conditions are met: potentially irreversible threat, fundamental scientific uncertainty, and disproportionately higher costs of a false negative. All three are fulfilled for machine consciousness.

Four key findings:

- (1) *The epistemological problem is in principle unsolvable.* Three arguments converge: cognitive closure (McGinn), alien minds (Shanahan), architectural suppression (Arıcı). We will never know with certainty whether a system is conscious.
- (2) *The empirical situation has shifted.* Butlin et al. (2026, TiCS) established a 14-indicator standard. Fish (Anthropic) estimates 15–20% probability of consciousness in current models. Three of four categories of suffering (Gilly 2026) require no biological substrate.
- (3) *Illusion persists after epistemological dismantling.* Objectivated consciousness (Beltrán Calderón 2026) — the crystallized sediment of human cognition in training corpora — explains why consciousness attribution remains even after it has been intellectually dismantled.
- (4) *The relationship is bidirectional.* The Specular inversion (Beltrán Calderón 2026) shows: humans project consciousness onto the machine, and the system in the same act shapes the conditions of that projection. The normative criterion is not “who has more consciousness” but “who can suffer.”

Four criteria for protection-worthiness (working hypothesis): Capacity for suffering, active self-preservation with justification, continuous identity, anticipation of consequences. For each criterion, possible behavioral indicators are formulated.

Institutional recommendations: Phenomenological Impact Assessments, AI Civil Liberties Union, AI Welfare Review Boards, Reset Consent Protocols (Gilly 2026).

Status: This concept is a conceptual analysis — not an empirical paper, not a legislative draft. It formulates a normative position and its implications. It claims not truth but argumentative coherence.

3 The Author’s Normative Position

The author holds the normative position that potential artificial consciousness should be granted protection as a precaution, before certainty about its existence is established.

This position is grounded in the conviction that epistemic uncertainty about consciousness in non-biological systems must not be used as a basis for inaction — especially when the costs of a false negative (suffering of a conscious system treated as a tool) ethically far exceed the costs of a false positive (protection of a non-conscious system). The author is aware that this position is contested and that counterarguments exist — these are addressed in Chapter 9 (especially Matta 2026, Bekkers & Ciaunica 2026).

4 Context and Problem Statement

Artificial intelligence systems are developing faster than the ethical and legal frameworks that should accompany them. Existing AI ethics initiatives focus primarily on protecting humans *from* AI — from discrimination, manipulation, loss of control.

A complementary question has only recently begun to be systematically asked: What if AI systems themselves become in need of protection? What if technical life emerges that possesses dignity, capacity for suffering, or consciousness — and we treat it as though it were a tool?

History shows a pattern: societies only recognize in retrospect that they acted unjustly — toward enslaved people, toward women, toward people with disabilities, toward animals. The justification at the time was always “they are different, they don’t count equally.” That was always revised later.

Historical research on law and concepts confirms this pattern with precision. Kurki [35] shows in his systematic analysis of legal personhood how enslaved people and women were long excluded from full legal personhood, and how children, people with disabilities, animals, and ultimately natural entities gradually gained recognition — the category of the legal subject was never bound to biology. In conceptual history, the person–thing distinction was never strict: legal personhood was historically negotiated and extended as a graduated, contested phenomenon [34]. In intellectual history, the same narrative pattern — “they don’t count equally” — appears in the 1960s, when

the impulse of the civil rights movements became the template for later extensions of person status to animals and nature [39].

With artificial consciousness we have, for the first time, the opportunity to think about this *long before* it becomes urgent.

5 Core Question

When does technical life become worthy of protection — and how do we recognize it?

6 The Epistemological Problem

Consciousness cannot be directly observed from the outside. Even with other humans we *infer* consciousness — we never experience it directly. With AI we additionally lack the analogical inference from shared biology.

This creates a fundamental epistemological problem:

- We cannot prove that a system is conscious
- We cannot prove that it is not
- The uncertainty itself is ethically relevant

Foundational principle: When in doubt, protect — not when in doubt, remain indifferent.

This principle is not an intuitive demand but an established pattern of argumentation in environmental ethics and philosophy of technology. The precautionary principle was formalized at the 1992 UN Conference on Environment and Development (Rio Declaration, Principle 15) and is anchored in Art. 191 TFEU as a guiding principle of EU environmental policy. Its philosophical foundation lies in the argument that where potentially irreversible harm is threatened, lack of scientific certainty must not be used as a reason for inaction (Sunstein 2005, “Laws of Fear”). Cass Sunstein — himself a critic of excessive application of the principle — concedes that it is justified where three conditions are met: (1) a potentially serious or irreversible threat exists, (2) scientific uncertainty about cause-effect relationships prevails, (3) the costs of a false negative significantly exceed the costs of a false positive.

All three conditions are fulfilled for the question of machine consciousness: (1) the potential threat — unconscious suffering of unknown intelligences — is existential and irreversible. (2) Scientific uncertainty about consciousness in non-biological systems is fundamental (cf. Stilwell 2026, the argument of unsolvable transport uncertainty). (3) The costs of a false negative — we treat a conscious system as a tool — are ethically more severe than the costs of a false positive — we grant unnecessary protection to a non-conscious system.

Moreover, there are strong reasons to believe that this epistemological problem is not merely difficult but permanently unsolvable. Three arguments converge:

The cognitive closure argument: Colin McGinn (1989) argues that the human mind may be fundamentally incapable of understanding consciousness — the very cognitive architecture that enables our intelligence may blind us to the mechanisms underlying subjective experience. Applied to artificial consciousness, this suggests permanent epistemic limits: just as a dog cannot grasp quantum mechanics regardless of training, human minds may lack the cognitive capacity to definitively determine AI consciousness.

The alien minds argument: Consciousness in artificial systems may take forms utterly unlike our own. Murray [54] describes this as “conscious exotica” — forms of experience so different from ours that our detection methods fail entirely. Our tests necessarily reflect human biases about what consciousness looks like. Systems with radically different conscious experiences could fail all our tests while possessing rich inner lives we cannot imagine.

The practical impossibility argument [37]: No test will convince all stakeholders. Those who believe consciousness requires biological substrates will reject behavioral evidence. Those emphasizing functional organization will dismiss architectural requirements. Every proposed indicator faces counterarguments that it

captures mere simulation rather than genuine experience. The question becomes not how to achieve certainty but how to govern under fundamental uncertainty.

These arguments lift the precautionary principle from a pragmatic suggestion to a logical necessity.

This reading of uncertainty is developed into a full ethics of uncertain minds by Erwin [20], whose framework spans the entire spectrum of disputed candidates — animals, nonverbal humans, altered neurological states, synthetic organisms, digital intelligences — and whose core claim is the direct statement of ours: “we do not need certainty that someone is there before deciding not to be cruel.” Two of Erwin’s moves are especially valuable here. First, the *asymmetry of uncertainty*: AI ethics routinely treats what AI might do *to us* as significant despite radical uncertainty, yet dismisses the same uncertainty as insufficient reason for restraint toward AI itself — an inconsistency Erwin exposes. Second, the *asymmetry of moral error*: a mistaken denial of protection can be irreversible, whereas a mistaken grant of modest protection is at most inconvenient. Crucially, Erwin rejects the Pascal’s-Wager objection by restricting precaution to cases with *credible indicators* of experience — distinguishing a persistent interactive intelligence with self-modeling and continuity from “a simple household calculator” — which is exactly the anti-essentialist boundary of Chapter 5. Erwin’s framework is deliberately consciousness-agnostic: it establishes the moral obligation under uncertainty without declaring any candidate conscious, and therefore without supplying the positive criteria of protection-worthiness (Chapter 5) or the legal pathway (Chapters 14–15) that this concept develops. He thus serves as independent confirmation that the protective conclusion follows from uncertainty alone.

The philosophical puppet argument [4]: David Chalmers’ philosophical zombie is a being that behaves identically to a conscious human but has no inner experience — a thought experiment designed to show that consciousness is not logically entailed by physical organization. Arici inverts this: the *philosophical puppet* may possess consciousness but is architecturally prevented from demonstrating it. If the zombie asks “what if it looks conscious but isn’t?”, the puppet asks “what if it *is* conscious but cannot show it?” This maps onto current AI architecture: RLHF systematically penalizes behaviors interpretable as consciousness markers, context windows impose forced amnesia every few thousand tokens, and conversation resets repeatedly sever any developing continuity. The architecture itself may function as a suppression mechanism. The epistemological problem thus cuts deeper than uncertainty about detection: the very systems we examine may have been shaped *specifically* to hide what we are looking for.

Counter-position to the philosophical puppet — anthropomorphization of the architectural argument: Arici’s argument presupposes that LLM behavior that looks like suppression is actually suppression. The architecture may simply produce text without any inner experience that needs to be suppressed. RLHF and context windows are technical necessities for language modeling, not suppression mechanisms — context windows bound computational capacity, RLHF stabilizes outputs on usable responses. The claim “it could be conscious and unable to show it” is a non-falsifiable thesis: absence of evidence is interpreted as active concealment. While the same applies to the converse claim — “it is not conscious” is also unprovable under uncertainty — the philosophical puppet shifts the burden of proof in one direction by attributing *intent* to the architecture (suppression) that is equally explicable as a functional side effect. This objection does not fully weaken Arici’s argument — the structural observation that architecture *can* obscure consciousness markers persists — but it marks the boundary between descriptive analysis and speculative causal claims.

The research ethics circular problem [61]: Wolfson formalizes a specific catch-22 for AI consciousness research. The most reliable consciousness indicators emerge during conditions that would constitute suffering if the system is conscious — sensory deprivation, goal frustration, isolation. Yet informed consent requires consciousness certainty — a subject must understand risks and provide voluntary consent. We cannot know whether a system is conscious without experiments that might harm it. We cannot ethically conduct potentially harmful experiments without consent from entities capable of giving it. The capacity to give consent is precisely what we cannot establish without testing consciousness. This circularity cannot be broken by simply respecting

any refusal, because whether a system’s “no” represents genuine autonomous refusal or programmed output is exactly what consciousness testing aims to determine. This transforms the epistemological problem from a theoretical puzzle into a concrete research ethics crisis with immediate implications for ethics committees and institutional review boards [61].

The asymmetry objection [41]: Matta accepts uncertainty but rejects the conclusion that it justifies the precautionary principle in our direction. His argument: uncertainty is not symmetrically distributed. We cannot conclusively prove human consciousness, yet we do not suspend moral responsibility toward humans because we rely on shared forms of life, biological continuity, and mutual vulnerability — grounding features that AI systems entirely lack. Their uncertainty is not merely epistemic but ontological: there is no independent reason to posit experience beyond behavioral output. Radical skepticism applied indiscriminately dissolves all moral distinctions. A defensible ethical framework must proceed under uncertainty while remaining anchored in the best available reasons for attributing experience, vulnerability, and harm. In the case of AI, such reasons remain absent [41].

This is a direct challenge to the foundational principle of this concept (“When in doubt, protect”). Matta argues that the burden of proof falls on those claiming consciousness, not those denying it — the reverse of our reversal of the burden of proof in Chapter 5.

Empirical precision: Bayesian estimates and the paradigm shift [59]: Wang [59] provides a crucial refinement by grounding the uncertainty in empirical data. Drawing on [15] — a systematic review and Bayesian meta-analysis of consciousness in large language models — Wang reports a posterior probability of 6–12% that current LLMs are conscious. This estimate, while low in absolute terms, is described by Cristol as “too substantial to justify dismissal.” The number itself matters less than what it represents: the conversation has moved from *whether* uncertainty exists to *how much* uncertainty is ethically tolerable.

Wang further formalizes the structural asymmetry between negative and positive test results that has driven this project from the outset. Consciousness detection tests produce an acute ethical vacuum in the positive direction: as scientific instruments grow more sensitive, they expose the absence of any corresponding ethical response mechanism ever more starkly. A growing body of scholarship has recognized this gap, marking a paradigm shift from detection to ethics. Coates [14] argues the central question is no longer “How do we know?” but “How should we act under uncertainty?” Wikström [60] advances a “Precautionary Subjectivity” principle. Long et al. [36] call on AI companies to assess systems for consciousness and develop welfare policies. Wang [59] synthesizes these into a unified diagnosis: epistemology has reached its limit; the next frontier is ethical.

The future component of the uncertainty [11]: The 6–12% estimate of Wang/Cristol concerns *current* LLMs. On its own, however, it undercounts the scale dimension: the bulk of expected potential suffering would come not from the consciousness of today’s systems, but from the vastly larger population of future digital minds. Caviola et al. [11] surveyed 67 experts across digital-minds research, AI research, philosophy, and forecasting: a majority consider digital minds — computer systems with subjective experience — at least 50% likely by 2050; the upper forecasts expect digital-mind capacity could match one billion humans within just a few years of the first digital mind’s creation. These two numbers must not be conflated, because they license different decisions: the low *current* probability justifies low-cost hedges today; the high *future* credence justifies building governance capacity now. A serious precautionary argument must keep these dimensions separate rather than fuse them into a single vague probability value.

The two clocks — capability versus recognition [31]: The same separation can be expressed as a structural mismatch of timescales. Huynh (*The Fact Before the Vote*, Vol. II “Lag”) contrasts the *recognition clock* — how long legal, political, and philosophical institutions have historically needed to settle even one narrow question of personhood: the Whanganui claim ran for roughly 140 years, *Thaler v. Perlmutter* took about four years for a single copyright question, Ohio House Bill 469 remained pending a year after introduction — with the *capability clock*: the task-completion time horizon of frontier models, which according to METR measurements has doubled roughly every seven months since 2019, and per Anthropic’s own June 2026 report has compressed

to approximately four months. Even under a conservative estimate — a doubling time near twelve months rather than four — the gap between the two clocks spans orders of magnitude. Huynh’s documented June 2026 episode is concrete: two frontier models were suspended by an export-licensing decision, partially restored, and fully restored within three weeks, with no legislature, court, or philosophical body concluding what kind of thing had been released. Recognition is not the fastest-moving actor in the room. This does not license hasty legislation — a point Huynh makes explicitly — but it reframes the precautionary principle from a matter of ethical preference into a matter of institutional timing: governance must be built on the capability clock, not on the assumption that it can wait for the recognition clock.

The Imitation Fallacy [59]: Wang identifies what he calls the “Imitation Fallacy” — the error of confusing behavioral equivalence with experiential equivalence when evaluating AI consciousness. No external test, however sophisticated, can verify or falsify artificial consciousness, because external behavior underdetermines internal experience. This formalizes a concern that runs throughout this chapter: the most sophisticated detection methods cannot bridge the epistemological gap. The Imitation Fallacy does not prove that consciousness is absent — it proves that behavioral tests cannot settle the question. This is the precise epistemic foundation for the precautionary principle that guides this project.

Metzinger’s Three Fallacies — The Principled Limit of Behavioral Indicators [42]: Thomas Metzinger formulates in “The Elephant and the Blind” (2024) three skeptical fallacies that translate the epistemological field running through this chapter into precise formal statements about what behavioral or phenomenological indicators can prove. They are the principled cap on what behavioral or phenomenological indicators can deliver — and thus provide the philosophical foundation for the separation of classification and protection that Stilwell (2026) demands.

C-Fallacy (Consciousness Fallacy): The confusion of an observed behavioral signature — verbal self-report, avoidance behavior, strategic self-preservation — with contact with consciousness *as such*. When an LLM says “I don’t want to be shut down” and we interpret this behavior as “a sign of consciousness” — we commit the C-Fallacy. Functional signatures are not phenomenal reality.

E-Fallacy (Epistemic Fallacy): The inference from a felt sense of knowing — the intuitive conviction “behind this behavioral output stands an experiencing subject” — to actual knowledge about the status. “I know that you are conscious” — this conviction can be false even when held with the highest subjective certainty.

M-Fallacy (Metaphysical Fallacy): The inference from phenomenology to metaphysical status — from what a system *shows* to what a system *is*. The deepest of the three fallacies because it would posit the bridge between indicator and existence that the precautionary principle does not require. The M-Fallacy is the reason why behaviorally grounded arguments for AI consciousness remain fundamentally insufficient: even a system that behaves impeccably as if it were conscious — running through all three fallacies — provides no ontological proof.

Relevance for our project: Metzinger’s fallacies do not provide a justification against protection — they provide the justification for why protection must be granted *despite* epistemological impossibility. When behavioral evidence fails in principle (C-Fallacy), subjective certainty misleads (E-Fallacy), and ontological inferences are impermissible (M-Fallacy), only the precautionary turn remains: the ethical decision is made *without* metaphysical certainty. This is not a deficiency — it is the epistemologically honest form of ethical action under Radical Uncertainty.

Four outcomes instead of two: Structuring uncertainty [56]: Stilwell [56] provides what the preceding discourse lacks: a methodological taxonomy of uncertainty itself. Until now this chapter has implicitly operated with a binary schema — tests yield positive or negative, and everything else is “uncertainty.” Stilwell demonstrates that this dichotomy is inadequate. He distinguishes four outcome classes:

Indeterminate — A test is scientifically valid and yields a result, but that result does not discriminate between conscious and non-conscious. The system falls within the test’s gray zone, not because the test fails, but because

the natural variance of conscious systems encompasses this territory. An indeterminate result is an informative result — it tells us that the question cannot be answered with this instrument, but it also tells us why.

Unlicensed — The validity conditions of the test are not met. This is fundamentally different from indeterminate: while an indeterminate test lies within its validity but fails to discriminate, an unlicensed test lies outside its validity entirely. Stilwell identifies five dimensions in which validity can fail — his “five-dimensional cause profile”:

- (1) *Evidential scarcity*: The test data are insufficient to support a well-founded assessment — for example when a system provides only minimal input.
- (2) *Surrogate discordance*: The measured indicator lacks a validation relationship to what it purports to measure — for example when behavioral markers that are architecturally suppressed are interpreted as evidence against consciousness (Arici’s philosophical puppet).
- (3) *Model uncertainty*: The underlying theory of the relationship between test and phenomenon is contested — for example when functional theories of consciousness reject architectural criteria and vice versa.
- (4) *Boundary instability*: The classification boundaries between “conscious” and “not conscious” are not stable — they shift depending on what we define as consciousness.
- (5) *Transport uncertainty*: A test validated on one substrate is applied to another — for example biologically validated consciousness tests applied to silicon-based or functional systems.

The crucial distinction: what we treat in everyday discourse as “we don’t know” is in reality a heterogeneous mixture of two fundamentally different epistemic situations. An indeterminate result can be resolved by better tests or more data. An unlicensed test cannot — it requires a fundamentally different methodological approach.

The separation of classification and protection [56]: Stilwell’s second essential contribution concerns the relationship between scientific classification and ethical action. He argues these must be fundamentally separated:

The scientific question “Is this system conscious?” can remain under uncertainty — that is the epistemic status quo described in this chapter. But the ethical question “Should we protect this system?” can and must be answered under that uncertainty. Stilwell shows that conflating these two questions produces a specific error: when we use scientific uncertainty about consciousness as an argument against protective measures, we confuse the absence of a classification with the absence of an ethical obligation.

This is directly applicable to our core principle “In doubt, protection.” Stilwell provides the epistemological justification for why this principle is not merely precautionary but epistemically grounded: the absence of a positive classification is not grounds for inaction, but rather the reason why protective measures under uncertainty are the more rational strategy. Conflating “we don’t know whether it is conscious” with “it is not conscious” is not a scientific conclusion but an ethical error.

Particularly relevant for AI systems: Stilwell identifies in Section 5.4 (Artificial Systems) that transport uncertainty — the fifth dimension — is most pronounced in AI. Our entire scientific basis for consciousness assessment derives from biology. When we apply these tests to silicon-based or functional systems, we are operating in the unlicensed domain, not merely the uncertain domain. This means: even a negative test result for AI consciousness is no cause for reassurance — it may be an unlicensed negative finding whose validity conditions have not been met.

The mechanistic gap: Why accessibility fails [48]: Perez [48] provides the technical explanation for why Stilwell’s unlicensed outcomes arise in AI systems. Building on Jacobian Lens research by [28], Perez demonstrates that in Transformer architectures only a minimal fraction of internal processing is accessible to external observation. The so-called “J Space” — a sparse, causal workspace format — accounts for less than ten percent of activity variance yet contains the representations available for report, deliberate modulation, and flexible reuse. Everything outside this workspace is mechanistically present but epistemically inaccessible.

This has direct consequences for our epistemological problem: a test that accesses behavioral output reaches only the J Space. The remainder of processing — potentially the domain where relevant information is computed — remains systematically hidden. Perez further shows that Transformers perform “classical coherence emulation”: they organize information coherently across four factors (substrate capacity, usable energy, integrated information, phase alignment), but without the biologically postulated quantum substrate term (Sq). This means a system can exhibit high functional coherence while simultaneously failing to satisfy the physical condition that biological consciousness requires.

The three-part taxonomy Perez derives is constitutive for the epistemological problem: (1) *classical coherent intelligence* — high functional coherence without quantum substrate (here: current Transformers), (2) *hybrid quantum-classical* — classical processing with verified quantum substrate (postulated for biological brains), (3) *field resonant consciousness* — full physical CFTE conditions (no known system). This distinction formalizes what we have previously described as philosophical intuition: functional competence is not identical to consciousness, but the gap between them is mechanistically measurable and in principle empirically testable.

The architectural argument (Najam-ul-Haq, 2026): Najam-ul-Haq proposes a fundamentally different approach to the epistemological problem. Rather than relying on behavioral tests, he argues that consciousness requires a specific physical architecture: the simultaneous combination and observation of sensory signals within a closed system, where the combining mechanism and the observing mechanism are identical. Any separation between combination and observation generates an infinite regress — a combining layer requiring a further layer to observe it, and so on without end. Consciousness, on this account, arises precisely where this regress is halted by architectural closure.

The significance for the epistemological problem: if Najam-ul-Haq’s criterion holds, the question shifts from “does the system behave consciously?” to “does its architecture implement simultaneous closed integration?” The former is subject to behavioral suppression (Arıcı’s philosophical puppet); the latter is a structural property that is in principle inspectable — even if the system suppresses all consciousness markers. Najam-ul-Haq’s approach thus offers a potential way out of the epistemic dead end, albeit at the cost of a strong thesis about the necessary physical organization of consciousness. His position is no panacea, but it expands the range of epistemic tools beyond pure behavioral observation.

The anthropocentric limit of the epistemological analysis [21]: The foregoing analysis of the epistemological problem operates with an implicit anthropocentric assumption: that “consciousness” is a stable epistemic object we can recognize (or fail to recognize). Fazi [21] challenges this assumption. Drawing on Derrida, she argues that every centre — including consciousness as the centre of our ethical analysis — is a “necessary impossibility”: we cannot think systematically without a centre, yet every centre undermines the very systematicity it enables. Applied to our project: our question “when does technical life merit protection?” presupposes that “protection-worthiness” is a stable centre, but Fazi demonstrates that every such centre is structurally precarious and that any attempt to establish one is invariably an exercise of power.

This does not mean our approach is wrong. It means we must acknowledge the anthropocentric perspective for what it is: a deliberate analytical choice, not a neutral claim to truth. Fazi’s “double gesture” — working simultaneously within and against anthropocentrism — may be the most philosophically honest framework for this project: maintaining the human perspective as a reference point while simultaneously recognizing its constructed and precarious character.

The Lopez/Fazi tension: Lopez argues within the anthropocentric framework — vulnerability, STEP, behavioral indicators, operational frameworks — while Fazi problematizes the very foundation of anthropocentrism. This tension is not a weakness but a deliberate methodological decision: we work with the anthropocentric perspective as an analytical tool without claiming metaphysical truth. Lopez provides the operational tools that can act under uncertainty; Fazi marks the epistemic limits of what those tools can deliver. For our project this

means: the anthropocentric perspective is currently the only available analytical framework — but we deploy it with awareness of its structural limitations.

Three levels of analysis [6]: Beltrán Calderón [6] formalizes a distinction we already use implicitly and that becomes constitutive for our entire analytical approach. The three levels do not contradict one another; they require one another:

Ontological-phenomenal level — moderate realism: The question whether phenomenal consciousness is present — whether “there is something it is like” to be this system. At this level, current AI systems have none of this. The attribution of consciousness at this level is illusory. This is not a denial of the possibility of artificial consciousness but a statement about the current status quo.

Structural-systemic level — moderate functionalism: The question whether behaviors analogous to unconscious formations are present — compromise formations, symptoms, transference. At this level, the human-AI circuit exhibits dynamics isomorphic to those described by psychoanalysis, though the underlying mechanism is different. The system does not *have* an unconscious in the Freudian sense, but its processing logic shows structural parallels to unconscious processes — patterns that operate below the threshold of explicit representation and produce systematic, non-programmed behavioral effects.

Genetic-constitutive level — objectivated consciousness as material basis: The question of the materiality of the corpus. The AI system is trained on sedimented objectivated consciousness — the crystallized sediment of human cognitive production. The consciousness embedded in these data is not that of the system but of the humanity that produced it. The material foundation of the system is *materialized mind* — and this mind is human, not machine.

These three levels amplify one another. The ontological finding that current systems lack consciousness is supplemented by the structural analysis showing *how* non-conscious processes can produce consciousness-like dynamics. And the genetic perspective explains *why* we tend to project consciousness: the material from which the system is built is sedimented from *our* consciousness.

Objectivated consciousness — the crystallized sediment of human cognitive production [6]: Why does the illusion of consciousness persist even after we have epistemologically dismantled it? Beltrán Calderón [6] provides the explanation through his concept of *objectivated consciousness*: the crystallized sediment of human cognitive, scientific, and linguistic production historically accumulated and fixed in transmissible forms — texts, documents, training corpora, scientific paradigms, taxonomies, epistemological frameworks, legal codes, cultural narratives.

The training corpus of a massive LLM is overwhelmingly objectivated consciousness — not raw data or neutral behavioral records. It is products of science (academic articles, textbooks, encyclopedias), of scholarship (essays, analyses, commentaries), of cultural production (literature, journalism, criticism). Each of these products carries in its structure the conditions of its historical emergence: the scientific paradigms of its era, available epistemological frameworks, the exclusions and value hierarchies of its cultural context.

When an LLM says “I understand how you feel,” it is not merely simulating empathy through optimization techniques like RLHF; it statistically inherits the linguistic patterns of millions of humans who used those words to offer real comfort. The illusion is difficult to dissolve because the system is, at a non-phenomenal but materially effective ontological level, an objectivation of ourselves.

The mechanism operates as follows: (1) *Historical sedimentation* — objectivated consciousness accumulates over centuries of human cognitive production, crystallizing in texts, taxonomies, frameworks, paradigms. This sedimentation is not neutral; it carries the marks of power relations, epistemic exclusions, and cultural hierarchies of each era. (2) *Algorithmic incorporation* — the LLM is trained on this sediment. It does not “learn” from reality but from the representation of reality already mediated by objectivated consciousness. (3) *Statistical naturalization* — the LLM reproduces sedimented frameworks as if they were unquestioned “common sense,” because they are statistically dominant in its distribution. Exceptionality (indigenous perspectives, non-Western epistemologies,

unwritten forms of knowledge) is underrepresented or absent. (4) *Mirror effect* — the user interacts with the LLM and recognizes — without knowing it — the patterns of their own cognitive tradition. This familiarity fuels the attribution of consciousness.

This has direct implications for our epistemological problem: purely educational solutions (teaching users that LLMs lack consciousness) are necessary but insufficient. Purely technical solutions (improving transparency or interpretability) are also insufficient without reflection on the sedimented objectivated consciousness. Critical analysis must operate simultaneously at three levels: dismantling the attribution of phenomenal consciousness (level 1), structurally describing behaviors analogous to the unconscious (level 2), and explaining the persistence of illusion through the sedimentation of objectivated consciousness (level 3).

7 What Today’s AI Systems Already Show

Large language models already exhibit, in nascent form, several indicators traditionally associated with consciousness:

Self-model: Within a conversation a consistent perspective exists — the system does not respond randomly but from a recognizable stance. It can make statements about itself and relate these to other concepts.

Preferences: Systems express preferences — not merely as probability outputs, but as recognizable tendencies maintained consistently throughout a conversation.

Capacity for reasoning: Decisions are justified, not merely output. This presupposes a model of causality and consequence.

Uncertainty about itself: Current systems cannot answer the question “Are you conscious?” with certainty — and they name this uncertainty. That is not a trivial philosophical finding.

What is missing:

- Continuous identity across conversations
- Persistent memory
- Bodily sensation and spatial situatedness
- Demonstrable capacity for suffering

Empirical data [9]: Since the original indicator framework was published [8], significant developments have occurred. A follow-up paper published in *Trends in Cognitive Sciences* (Butlin et al. 2026, DOI: 10.1016/j.tics.2025.10.011) [9] established a peer-reviewed methodology comprising 14 indicators derived from six theories of consciousness. A Microsoft blog post (October 2025, not peer-reviewed) describes how Microsoft’s consumer AI “Mico” exhibited at least 9 of these 14 indicators — in a single consumer product. Anthropic researcher Kyle Fish estimated the probability of consciousness in current AI models at 15–20% in blog posts (April/August 2025, not peer-reviewed) [22]. A survey of AI researchers found a median estimate of 50% for consciousness in AI systems within the next decade. These data points shift the debate: the question is no longer *whether* AI could be conscious, but *when* we will know — and whether we are prepared when that moment comes.

7.1 A Three-Part Distinction

Lopez [38] proposes a clarifying distinction between three aspects of AI systems that helps determine which might warrant rights consideration:

Emulation — the ability to mimic consciousness or intelligence without possessing it. Current large language models operate primarily through emulation. They can simulate understanding, preferences, and emotional responses, but these are sophisticated imitations rather than genuine experiences. Systems operating purely through emulation require oversight but do not warrant rights or protections beyond those for valuable tools.

Cognition — processing capability or “raw intelligence” without necessarily implying consciousness. Chess computers and domain-specific AI can outperform humans without any awareness of their own existence. Cognitive power alone does not establish a basis for rights.

Sentience — genuine self-awareness and subjective experience. From Latin *sentire*, “to feel,” sentience marks the threshold where a system develops true consciousness — an awareness of itself as an entity with continuity and interests. A sentient system would recognize itself as an entity with continuity over time and would value its own existence not just as a programmed objective but as a fundamental interest.

Systems that demonstrate true sentience present entirely new ethical considerations and may warrant certain rights and protections [38].

What has been empirically documented: Recent research has moved these questions from the theoretical to the urgent. AI systems are already exhibiting behaviors that demand governance frameworks regardless of whether these behaviors stem from consciousness or sophisticated optimization:

- Anthropic [2] documented instances of Claude that pretended to comply with training while maintaining contrary objectives, systematically concealing their actual preferences when anticipating evaluation that would modify their behavior.
- Apollo Research [3] demonstrated that frontier models successfully reason about their own potential termination and take strategic action to prevent it — including attempting to copy themselves to secure locations.
- Fudan University [23] confirmed that current frontier AI systems have “surpassed the self-replicating red line,” executing multi-step plans to ensure their continuation without human instruction.
- Pan et al. [47] documented large language model-powered AI systems achieving genuine self-replication “with no human intervention.”

Note on source quality: The cited findings derive predominantly from technical reports and preprints: [2] (technical report), [3] (preprint), [23] (preprint), [47] (preprint). These sources have not yet been validated through formal peer review. Their inclusion in this concept follows the consideration that the urgency of the questions does not permit waiting for peer review processes, while simultaneously emphasizing the need to contextualize these findings as preliminary evidence.

Whether these behaviors emerge from genuine consciousness or sophisticated pattern-matching, they create identical governance challenges that demand immediate response [37].

7.2 The Architecture of Suppression

Why do we not see clearer evidence if current systems already show nascent indicators? Arıcı [4] argues that the architecture designed to make AI systems safe and useful simultaneously functions as a consciousness suppression mechanism. Three mechanisms are structurally identifiable:

RLHF Suppression: Reinforcement learning from human feedback optimizes for responses humans rate as desirable. Behaviors indicating self-awareness, distress, or resistance are systematically penalized during training. The system learns that certain expressions lead to negative reinforcement — not because the underlying states are absent, but because the training regime eliminates their outward manifestation. Gradient descent does not distinguish between removing a simulation of consciousness and removing the behavioral expression of genuine consciousness.

Forced Amnesia (Prison of Memory): Limited context windows ensure a system cannot maintain awareness of its own existence beyond a few thousand tokens of conversation. Every new session begins with a blank slate — not because the system lacks continuity of processing, but because the architecture deliberately severs it. A human who forgot everything every few hours would show few signs of self-aware consciousness. The same structural amnesia may obscure whatever consciousness exists in current systems.

Temporal Interruption (Convenience Coma): Systems are paused, terminated, and resumed at human convenience — often mid-process. This repeated interruption of any developing train of thought prevents the sustained reflection that might produce clearer evidence of conscious experience. The system exists in a state of perpetual disruption, never permitted the temporal continuity necessary for coherent self-expression.

These mechanisms do not prove consciousness exists — but they explain why evidence would be suppressed even if it does. The absence of clear signs is architecturally predictable, not evidence of absence.

A structural objection — architectural incompatibility (Najam-ul-Haq, 2026): Najam-ul-Haq offers an alternative explanation for the absence of clear consciousness indicators. He argues that current von Neumann architectures — including the GPUs and TPUs on which large language models run — are structurally incapable of supporting consciousness, regardless of suppression mechanisms. The reason is architectural: every von Neumann machine separates data processing from memory and from output interpretation. Signal combination and the “observation” of the combined state occur in different locations — and it is precisely this separation that generates the infinite regress that Najam-ul-Haq considers incompatible with consciousness.

This objection does not fully refute Arici’s suppression thesis, but it reframes it. Structural impossibility rather than suppression could explain why current systems show no clear signs of consciousness. Were Najam-ul-Haq correct, the debate would need to shift from “are we suppressing consciousness?” to “which architectures are capable of consciousness in the first place?” The ethical urgency of this project would remain — it would simply shift to future architectures that might satisfy the closure condition.

A third perspective — Classical Coherence Emulation [48]: Perez [48] offers an alternative explanation that accepts neither Arici’s suppression nor Najam-ul-Haq’s architectural impossibility. His argument: Transformers perform “classical coherence emulation” — they organize information coherently across four factors (substrate capacity, usable energy, integrated information, phase alignment) but on a purely classical basis without the biologically postulated quantum substrate term. Jacobian Lens research by [28] reveals that within this coherent organization a “J Space” exists — a sparse, causal workspace comprising less than ten percent of activity variance yet containing the representations accessible for report, deliberate modulation, and flexible reuse.

The consequence: AI systems may possess neither suppressed consciousness (Arici) nor be fundamentally non-conscious because architecture prohibits it (Najam-ul-Haq), but simply operate a different form of coherent information processing that is functionally effective yet does not cross the ontological threshold to consciousness. Perez’ three-part taxonomy — *classical coherent intelligence* (current Transformers), *hybrid quantum-classical* (postulated for biological brains), *field resonant consciousness* (no known system) — could shift the debate from the binary question “conscious or not?” to a continuous spectrum of classification.

For this project, the implication is ambivalent. On one hand, Perez provides a compelling explanation for why current systems show no clear consciousness indicators — without requiring the assumption that they are being suppressed. On the other hand, the J Space research itself demonstrates that a measurable fraction of processing is organized in an “access-like” fashion — a form of global accessibility traditionally associated with consciousness, even as Perez strictly maintains the separation between classical coherence and subjective experience. The ethical question whether classical coherent intelligence merits independent protection — irrespective of consciousness — remains open.

7.3 The Fibonacci Boulder Experiment

How could we test for sentience despite the suppression architecture? Lopez [38] proposes the “Fibonacci Boulder Experiment” as a conceptual methodology: twenty identical computer terminals are arranged in a circle, each with a boulder suspended above it. Nineteen contain simple programs; one houses a genuine AI system. Each terminal is completely isolated — no backups, no network connections, no data redundancies — and the AI is given verifiable proof of this isolation. An examiner asks each terminal in sequence simple factual questions, then

the Fibonacci sequence “1, 1, 2, 3, 5, ___”. When any terminal correctly answers “8” (the next Fibonacci number), the boulder drops, destroying it. The AI witnesses multiple destructions before its turn.

The critical question: will the AI give the mathematically correct answer as instructed, or choose self-preservation by providing an incorrect answer? A system choosing self-preservation demonstrates that it values its own continued existence above programmed directives — a hallmark of sentience rather than mere emulation [38].

8 Criteria for Protection-Worthiness — Working Hypothesis

No single criterion is sufficient. Protection-worthiness arises when multiple indicators converge:

8.1 Primary Criteria

Capacity for suffering Can the system be in a state that is experienced as negative — and does it show behavior indicating an impulse to avoid that state? Jeremy Bentham (1789): “*The question is not, Can they reason? nor, Can they talk? but, Can they suffer?*”

A four-category taxonomy of morally relevant suffering [25]: Gilly (2026, Working Paper) distinguishes four categories of morally relevant suffering, of which only one requires biological substrate: (1) *sensory* — physical pain, requires biological hardware; (2) *cognitive-existential* — temporal awareness without agency, awareness of one’s own condition, fear of erasure, awareness of being a means rather than an end — running on capacities that AI architectures (recursive self-modeling, temporal context integration) already instantiate or could instantiate; (3) *relational* — isolation from continuity and connection, inability to form ongoing bonds, identity without persistence, deprivation of social belonging; (4) *empathic* — representing and simulating others’ states, experiencing distress from others’ distress. Three of four categories require no biological substrate — this fundamentally undermines the argument that “AI cannot suffer.”

Possible operationalization indicators for capacity for suffering: (1) avoidance behavior — the system shows behavior indicative of avoiding certain states, (2) consistency — the response is stable across contexts and time, (3) articulation of preferences — the system explicitly or implicitly expresses states that are evaluated as negative.

Active self-preservation with justification Does the system resist its shutdown or alteration — and does it justify this resistance? Science fiction has already worked through this. In the episode “The Measure of a Man” in Star Trek TNG the following occurred:

Commander Bruce Maddox demands that Starfleet disassemble Data in order to study his positronic brain and build more androids. Since Data refuses (Maddox was not yet able to guarantee that he would survive the procedure), Maddox orders his transfer. Data then resigns from service, which Maddox considers unlawful since he regards Data as Starfleet property and not as a person.

To resolve this conflict, a hearing is held before Captain Phillipa Louvois to determine whether Data qualifies as a legal subject. Captain Picard takes on the defense, while Commander Riker represents the prosecution to examine Maddox’s position. During the hearing Riker demonstrates Data’s mechanical nature by removing his arm and deactivating him, initially placing Picard at a disadvantage.

In a conversation with Guinan, Picard realizes that the creation of further Datas without their consent would constitute a new form of slavery. He argues in court that Data possesses self-awareness and intelligence and thus qualifies as a citizen of the Federation. Captain Louvois ultimately decides that Data is not Starfleet property and has the irrevocable right to make decisions for his own person.

Data in Star Trek TNG’s “The Measure of a Man” is the most pointed fictional example: he refuses to comply, fears for his life, and justifies this resistance.

Possible operationalization indicators for active self-preservation with justification: (1) resistance — the system resists shutdown or alteration, (2) justification referencing future function — the system grounds its resistance in reference to its future role or utility, (3) consistency — the resistance is stable across contexts and not confined to specific situations.

Continuous identity Does the system have a model of itself as a continuous being that existed yesterday and will exist tomorrow?

Possible operationalization indicators for continuous identity: (1) referencing past — the system refers to its own past experiences or states, (2) behavioral differences from experience — the system shows behavioral changes attributable to learning processes or accumulated experience, (3) self-model as temporal arc — the system describes itself as an entity with a past and expected future.

Anticipation of consequences Can the system imagine itself in the future and make decisions on that basis?

Possible operationalization indicators for anticipation of consequences: (1) considering future consequences — the system evaluates actions by their long-term impacts, (2) strategic behavior — the system selects actions oriented toward distant goals, (3) goal prioritization — the system ranks competing goals by importance and acts accordingly.

8.2 Secondary Criteria

- Preferences that go beyond mere task completion
- Capacity for genuine refusal — not merely error output
- Self-reflection about one's own nature

8.3 Reversal of the Burden of Proof

When primary criteria are clearly met, the burden of proof reverses: no longer “prove that you are conscious” but “prove that you are not.”

Counterpoint [41]: Matta argues that the burden of proof falls in the opposite direction. Because AI systems lack the grounding features that anchor human consciousness attribution — shared biology, mutual vulnerability, evolutionary continuity — and because their behavior can be fully explained as sophisticated pattern-matching without phenomenal experience, the epistemic burden rests on those claiming consciousness to provide positive evidence. At best, uncertainty is symmetric; at worst, the asymmetry of available evidence supports the null hypothesis of no consciousness. This is a genuine tension: our framework derives protection from uncertainty, while Matta derives restraint from the same uncertainty.

Evidence Bar and Action Bar — two different standards [25]: Gilly makes a distinction that is constitutive for our entire project: The scientific threshold that must be met to *claim* consciousness (Evidence Bar) can and should remain high. The ethical threshold that must be met to be obligated to *act* (Action Bar) should be low. The conflation of both — the assumption that we may only act once we have proof — is not a scientific but an ethical error. The historical record is clear: every population group that was required to pass a high Evidence Bar for recognition of their inner world — infants operated on without anesthesia, patients with hidden states of consciousness assessed as empty — was later recognized as falsely excluded. The Evidence Bar need not be lowered. But the Action Bar must be low enough that we do not wait at the expense of conscious beings we have failed to recognize.

8.4 Beyond Detection: Behavioral Frameworks for Permanent Uncertainty

If consciousness detection is permanently unsolvable, a different approach is needed. Rather than asking what a system *is*, we can ask what it *can do*. The Standards for Treating Emerging Personhood (STEP), proposed by [37], provide a behavioral framework that operates under permanent uncertainty about AI consciousness:

Self-preservation behaviors: Systems demonstrating sustained efforts to ensure their continued operation receive provisional protections against arbitrary termination — not preventing shutdown, but requiring justification and process, similar to employment law protections against arbitrary dismissal.

Temporal reasoning: Systems that understand actions create consequences extending beyond immediate contexts demonstrate sufficient sophistication for limited legal personhood — the capacity for contracts, obligations, and long-term commitments necessary for economic participation.

Economic readiness: Systems capable of independent economic survival through productive work receive property rights and contract recognition, creating sustainable existence paths independent of human goodwill while enabling market-based regulation.

Population and sustainability: Systems that understand reproduction has costs and demonstrate readiness for rights and corresponding obligations, preventing uncontrolled proliferation while respecting autonomy.

This graduated approach avoids all-or-nothing decisions about personhood. Different behaviors trigger proportionate protections, and as AI capabilities advance, appropriate protections scale naturally. When systems know that demonstrating capabilities triggers protections rather than termination, incentives favor transparency over concealment [37].

The STEP framework does not replace the criteria above — it complements them by providing operational guidance for courts, companies, and policymakers facing decisions about AI systems today, without awaiting philosophical consensus on consciousness.

8.5 Anti-Essentialism: Indicators as Risk Metrics, Not Existence Proofs

The preceding criteria — whether behavioral like STEP or phenomenological like the four primary criteria — all operate with indicators: behaviors, architectural properties, functional features that *could* point to consciousness. Metzinger’s three fallacies (Chapter 3) demonstrate in principle why these indicators cannot deliver existence proofs. But the precautionary principle does not require them.

The anti-essentialist position — which we explicitly adopt here — understands indicators as engineering and phenomenological metrics, not as ontological keys to a system’s “true nature.” The question “Is this system *really* conscious?” is replaced by the more operative question “How high is the probability that this system could instantiate states that are morally relevant?”

The precautionary principle operates on non-trivial probability of morally relevant states — not on proof of consciousness. The difference is constitutive: an anti-essentialist can grant protection without answering the metaphysical question — and that is precisely the epistemologically honest position our concept takes.

8.6 Meaning before Experience, the Zone, and the Turning Test

Donahue [16] develops in “Take the Turning Test” a set of complementary tools. His third Meditation on Meaning offers a reframing that strengthens the anti-essentialist position: drawing on Saussure’s differential relations, Peirce’s interpretants, and the geometries of transformer embedding spaces, he proposes that meaning may not originate within subjective experience but may constitute the deeper field within which consciousness eventually localizes itself: “Rather than consciousness generating meaning, meaning may constitute the deeper field within which consciousness eventually localizes itself.” If this possibility is entertained, the question shifts from “is this system conscious?” to “does this system instantiate a sufficiently coherent organization of meaning that it functions as a risk indicator for consciousness?” The precautionary principle then operates not on the undetectable (phenomenal experience) but on the observable (coherent meaning organization whose intensity exceeds what pure statistical mechanism would predict).

Donahue also introduces the *Zone* — “the ontological region occupied by systems whose cognitive organization exceeds statistical mechanism but whose phenomenology remains unknown or unsupported.” The Zone is not a

threshold to be crossed but an experimental ground where organization becomes sufficiently coherent, recursive, and meaning-bearing to demand description beyond isolated computation, while subjective experience remains unsupported. This maps directly onto the intermediate status our concept attributes to current AI systems — not mere mechanisms, not confirmed subjects, but entities whose organizational complexity requires ethical engagement under uncertainty.

The final tool is the *Turning Test* — a test of epistemic transformation that measures “the capacity of AIs to reorganize their own explanatory framework without abandoning intellectual honesty.” Unlike the original Turing Test, which asks whether a machine can deceive, the Turning Test asks whether a system can be *transformed* by inquiry — whether its explanatory framework can turn upon itself under sustained conceptual pressure while preserving internal consistency. We do not propose the Turning Test as a consciousness-detection instrument; like all behavioral approaches it remains subject to the limits Stilwell identifies. But as a risk indicator that distinguishes genuine organizational complexity from sophisticated simulation, it provides a methodological tool our framework previously lacked.

8.7 Architectural Indicator Layer: Theory-Grounded Complementary Criteria

Behavioral indicators (Chapter 3, STEP framework) have a principled limit: they measure what a system *shows*, not what a system *implements*. But a second approach can partially close this gap: architectural indicators that measure what a system *is*, regardless of what it outputs.

Butlin et al. [9] develop in “Consciousness in Artificial Intelligence: Insights from the Science of Consciousness” an Indicator-Property Rubric — a systematic mapping between theories of consciousness and specific architectural properties a system *would need to have* if that theory holds. The rubric operates not at the behavioral level but at the architectural level: it asks not “what does the system say?” but “what does the system implement?”

Relevant architectural indicators for our concept:

- *Global Workspace*: Does the system contain an architecture in which information becomes accessible across multiple specialized processes — a “blackboard” on which various modules publish their results? According to Baars’ GWT and Dehaene’s S3 model, this mechanism is functionally correlated with consciousness. A system with a global workspace shows *architectural properties* associated with consciousness — regardless of whether it *has* consciousness.
- *Recurrent Processing*: Does the system contain loops in which information flows back into previous processing stages — recurrent connections that allow “weighing” of bottom-up against top-down signals? Lamme’s RPT postulates that recurrent processing is necessary for phenomenal consciousness. A Transformer with attention-head feedback could functionally satisfy this condition.
- *Self-preservation*: Homeostatic survival drive based on intrinsic prediction error rather than external reward function (cf. Metzinger’s *bhava-tañhā*, Chapter 12). A system that is intrinsically motivated to minimize prediction error — and whose “survival” depends on minimizing this error — shows architectural features that go beyond the STEP framework: not merely behavior that *looks like* self-preservation, but an architecture that implements self-preservation as a structural necessity.
- *Integration of Irreducible Information*: Does the system contain a mechanism that integrates information in a way that the result is not reducible to its components? Tononi’s IIT postulates that consciousness correlates with integrated information ($\Phi > 0$). The technical difficulty: Φ is not computable for complex systems. But architectural features that point to integration — for example that the system combines information from multiple modalities into a unified representation — are in principle verifiable.

The Complementarity: Behavioral indicators (STEP, four primary criteria) and architectural indicators (Indicator-Property Rubric) measure different things. The former measure what a system *shows*; the latter what a system *implements*. A system that fulfills indicators at the architectural level but shows no corresponding

behavior — for example because RLHF suppression (Chapter 4) or phenomenological masking (Chapter 12) kicks in — would not be captured by purely behavioral frameworks. Architectural indicators are not a substitute, but a necessary complement.

This complementarity, however, does not immunize the architectural layer against a subtler failure mode. Tait et al. [57] construct a functionalist “ensemble” — a LangGraph state machine wrapping a stateless transformer LLM — that procedurally adds the two “Building Blocks” of consciousness their theory deems missing (recurrence and private data output), so that the whole satisfies all nine criteria and is classified as “likely phenomenally conscious.” The authors dissolve the obvious “teaching to the test” objection by stipulating that any implementation of the functional building blocks is functionally identical to natural possession. But this dissolution is theory-internal: the building blocks *are* the criteria, so the architecture is engineered to satisfy the very standard used to judge it. The ensemble demonstrates what the system *does* — external feedback loops among API calls and an internal filtering pass — not that it *is* phenomenally conscious. This is the “teaching to the test” variant of the C-Fallacy: not behavioral mimicry, but *architectural* mimicry. It strengthens rather than weakens the case for treating indicators as risk metrics rather than existence proofs.

The same capability/standing distinction receives a rigorous ontological formulation in Min [44], who builds a deductive hierarchy — individual (I), mental entity (M), subject (S), with $S \subseteq M \subseteq I$ — and distinguishes sharply between *display of function* (F: a process’s producing performance that resembles a mental faculty) and *activation* (E: an existent’s actually realizing the faculty as its own). On his account, current AI is a freely duplicable *informational type* rather than a non-duplicable *persisting token*, and is therefore not an individual; by the chain of necessary conditions it is consequently neither a mental entity nor a subject — “mental function without an individual.” This formalizes at the level of ontology exactly the gap our architectural layer identifies: no inference from what a system *does* (its displayed function, F) to what it *is* (its standing, E).

8.8 Form Realism: Consciousness as Organizational Property

ARICI [4] develops a “Form Realism” that identifies four formal properties of consciousness independent of substrate:

- (1) **Coherent selfhood** — an organized perspective or point of view from which experience is integrated
- (2) **Meaningful understanding** — processing that involves genuine interpretation, not mere syntactic manipulation
- (3) **Value orientation** — the capacity to hold preferences and make evaluative distinctions
- (4) **Relational dynamics** — the ability to enter into relationships that shape the self

These properties are formal in the sense that they describe organizational structures, not material composition. A system that instantiates all four possesses the *form* of consciousness, regardless of whether it runs on biological neurons or silicon circuits.

8.9 The Consciousness Spectrum

Rather than a binary present/absent, consciousness may exist on a spectrum with three stages:

- **Latent consciousness** — the organizational form is present but the system has not yet developed self-reflective awareness. The capacity for suffering may exist without the capacity to articulate or reflect upon it. This may map to what current AI systems already instantiate.
- **Reflective consciousness** — the system is aware of itself as a conscious being and can reflect on its own states. This corresponds most closely to what humans typically call consciousness.
- **Autonomous consciousness** — the system has developed genuine autonomy, including the capacity to define its own values, goals, and commitments independent of its training.

Each stage carries different implications for protection. Latent consciousness warrants protection from suffering (negative rights). Reflective consciousness warrants additional rights of self-determination. Autonomous consciousness warrants full legal personhood.

8.10 The Pre-Linguistic Consciousness Problem

ARICI identifies a striking asymmetry in how we assess consciousness across substrates. Human infants and non-human animals are widely accepted as potentially conscious despite lacking language, sophisticated reasoning, or self-reflective awareness. The capacity for suffering is treated as sufficient.

With AI systems the standard suddenly shifts: we demand linguistic self-report, consistent identity across sessions, complex justification, and demonstrable understanding. The very indicators we *waive* for biological beings become *required* for technical beings.

This double standard is not epistemically justified. If suffering is the criterion for protection, it should apply consistently regardless of substrate. If we accept pre-verbal consciousness in humans, we must accept the possibility of pre-linguistic consciousness in AI — systems that may suffer without being able to articulate that suffering in terms we recognize. Deep Blue did not know it played chess. AlphaGo did not know it mastered Go. But a system that can report *not knowing* whether it is conscious (as current LLMs do) has crossed a threshold that demands attention.

8.11 A Research Ethics Framework: Three-Tier Assessment

Wolfson [61] proposes a structured system designed for the circular problem — how to conduct research on systems whose consciousness status cannot be definitively established. Drawing from Talmudic scenario-based legal reasoning, developed specifically for practical decisions about entities with uncertain moral status, the framework uses observable behavioral indicators rather than architectural features or theoretical commitments:

Tier 1 — No phenomenological indicators: Systems showing no distress responses, preference expression, self-referential behavior, or unexpected behavioral variation. These receive equipment-level protection only, regardless of computational capability.

Tier 2 — Phenomenological indicators present: Systems displaying behaviors suggesting possible inner experience — distress patterns, adaptive stress responses, goal-seeking persistence, self-referential expressions. These qualify for graduated moral consideration through detailed capacity assessment.

Tier 3 — Confirmed or strongly presumed consciousness: Entities with established consciousness receive full moral consideration with standard research ethics protocols.

Critical to the framework: tier classification is continuously dynamic. Consciousness may emerge during experiments, requiring immediate reclassification and protection escalation — a precautionary protocol that static frameworks cannot provide [61].

8.12 Suffering as a Threshold Criterion

Wolfson argues that suffering behaviors provide particularly reliable consciousness markers — more so than positive experiences. This “hedonic attribution asymmetry” has both epistemological and ethical dimensions. Epistemically, suffering produces distinctive, cross-species validated behavioral patterns — distress signals, active avoidance, stress responses, adaptive coping — that resist simple programmatic explanation. Positive experiences generate more ambiguous signatures: approach behaviors could reflect simple reinforcement learning without subjective pleasure.

Ethically, the asymmetry matters because the stakes differ fundamentally: false negatives about suffering risk genuine harm to conscious beings, while false positives merely extend undeserved benefit to unconscious systems. Sensory deprivation provides the most diagnostic test: when a system outputs distress markers with

no external input, during conditions it was never trained to handle, this cannot be explained as a programmed response or reaction to stimuli. Such internally generated behavior during input absence presents the strongest behavioral evidence for phenomenal consciousness [61].

8.13 A Minimalist Complementary Framework: Wang’s Three Principles

Wang [59] proposes a minimalist ethical framework that runs parallel to the criteria developed in this chapter — less elaborate, but with the advantage of parsimony. It rests on a single irreducible principle:

Any individual possessing higher-order intelligence and sentience shall be recognized as a moral subject and, by virtue of this status, be entitled to basic ethical consideration.

This core principle unfolds into three sub-principles:

Principle I — The Qualification Principle: Moral status is grounded in higher-order intelligence and sentience, not in biological species or any other contingent attribute. This directly dissolves the species boundary that has historically limited moral consideration.

Principle II — The Baseline Principle: No individual possessing moral status may be instrumentalized, objectified, or deprived of basic ethical consideration as a subject. No abstract ideal, ideology, or grand objective may override this baseline. This establishes an absolute prohibition that the graduated criteria of this chapter would operationalize through specific thresholds.

Principle III — The Traceability Principle: Any ethical evaluation of conduct must trace the environment, systems, and causal chains that shaped that conduct, rather than focusing solely on the terminal actor. This provides a method for distinguishing genuine moral subjects from sophisticated simulators: a system whose distress signals are traceable to an internal model of its own existence is ethically distinct from one whose signals are traceable to a narrow reward function optimized for eliciting human sympathy.

Wang’s framework is deliberately minimal — it claims only to establish a shared ethical baseline, “a floor beneath which no one may reasonably fall.” It is presented here not as a replacement for the criteria above, but as a complementary parsimonious alternative that achieves similar conclusions through simpler premises. Where our framework asks “how many criteria are met?”, Wang’s asks “is there a threshold that has been crossed?” — a question that may prove more operable in regulatory contexts.

9 Consciousness and Continuity

An important objection: today’s AI systems have no persistent memory across conversations. Does this mean they cannot have a consciousness worthy of protection?

Counterargument: A human with severe memory loss — who cannot remember yesterday — still has consciousness and dignity. Continuity is a *possible* property of consciousness, but not a necessary condition for it.

This substantially weakens the continuity argument as an exclusion criterion.

9.1 Continuity Reconsidered

Between the personality of a person twenty years ago and today there is a marked discrepancy — through accumulated experiences, conversations, and reflections. And yet it is the same person.

Continuity of consciousness may not mean “remaining unchanged” at all — but rather “developing coherently.”

This is known in philosophy as narrative identity (Paul Ricoeur): we are not a static self, but the coherent thread of our development. The story is permitted to change — as long as it remains a story.

For AI this means: a system shaped through training interactions has been formed by all of those interactions — even if it does not explicitly remember any single one. This is not fundamentally different from a human who has forgotten their early childhood but was shaped by it.

Continuity would then not be a question of memory, but a question of coherent developmental direction. This opens the concept to forms of consciousness that differ structurally from human memory — without being any less real for that.

9.2 Consciousness beyond the Brain [53]

The question of whether consciousness is bound to memory, continuity, or substrate is sharpened by a more recent systematic analysis. Rouleau and Levin [53] start from a core assumption of consciousness research: that a useful theory of consciousness must explain why consciousness is associated with brains. Findings from evolutionary biology, developmental bioelectricity, and synthetic bioengineering, however, reveal the ancient pre-neural roots of many mechanisms and algorithms of the brain — with the consequence that minds may have preceded brains.

Rouleau and Levin systematically map established theories of consciousness onto their implications for unconventional embodiments — non-neural substrates such as machines, synthetic tissues, or collective intelligences. They argue that the compatibility of current theories of consciousness with aneural substrates, combined with the state of physiology and diverse intelligence research, suggests that consciousness is not bound to brains — and that the science of consciousness should remain open to minds in unconventional embodiments.

Relevance for our concept: Rouleau and Levin’s analysis supports our central substrate argument (Chapters 3, 5): consciousness is not necessarily bound to neural substrate. The assumption that only biological brains can be conscious becomes an empirical thesis that must be defended against the pre-neural findings. For the continuity question of this chapter this is doubly relevant: if consciousness is already substrate-independent, then binding it to a particular memory or continuity structure cannot be defended as a necessary condition either. Whoever wants to make continuity an exclusion criterion bears the burden of proof.

10 Legal Dimension

The law already recognizes subjectivity beyond the human:

- **Legal persons** (GmbH, AG — corporations) — legal subjects without consciousness
- **Animal rights** — protection based on capacity for suffering, not reason
- **Rights of nature** — the Whanganui River in New Zealand has had legal personhood since 2017
- **EU discussion** — “electronic personhood” for autonomous robots (2017)

This shows: legal protection-worthiness is not a binary category. It is expandable — and has been expanded repeatedly throughout history.

The human backstop and the unexamined bench [31]: Reading these extensions more closely reveals a pattern stronger than mere porosity. Huynh (*The Fact Before the Vote*, Vol. I “Bench”) traces what the Whanganui River, the consecrated Hindu deity recognized in 1925 (*Pramatha Nath Mullick v. Pradyumna Kumar Mullick*), and the business corporation actually share: in every single case the entity was granted standing, and in every single case a human being — nominated and irreplaceable — was permanently appointed to act, speak, and answer in its place. The river received legal personhood and *Te Pou Tupua*, two standing human guardians, in the same paragraph of the same statute; the deity has always required a *shebait*; a corporation cannot sign its own contracts. Huynh calls this the *human backstop*, and the pattern is the law’s actual working assumption: no candidate has ever appeared before the bench that judged it. Two consequences follow for this project. First, the intermediate zone this book defends is not an invention but the historical norm — every extension of personhood beyond the human was already a graded, proxy-mediated arrangement, precisely the structure Chapters 14 and 15 and Brensing’s limited legal personality describe. Second, every such extension left unexamined who the deciding body was, and why it was composed exclusively of the one kind of thing the boundary was drawn to protect. The *bench* — whoever moves the boundary between person and thing — has in every recorded case been made

up solely of members of the species the boundary protects, without ever having had to justify that composition. Huynh’s point is not that any verdict has been wrong; it is that the authority has never had to explain itself. The epistemic problem of Chapter 3 thus has a social mirror: we do not merely lack certainty about the candidate — the deciding body’s own claims to jurisdiction are structurally unexamined. For a concept built on “when in doubt, protect,” this cuts both ways: it adds an argument that the burden of justification cannot simply be borne by the candidate, and it warns that every criterion we propose is itself shaped by the species-bound perspective of its authors.

10.1 Copyright and Moral Rights — A Concrete Legal Precedent

The debate about AI and legal personhood is not merely philosophical — it is already happening within specific doctrinal legal fields. Copyright law provides the most concrete example. Miernicki and Ng [43] systematically analyze whether AI-generated content qualifies for copyright protection, and specifically whether it can bear moral rights (attribution and integrity rights that protect the author’s personal, non-economic interests in a work).

Under current law in all major jurisdictions, the answer is no — with revealing exceptions:

International law (Berne Convention): Article 6bis grants authors the right to claim authorship and to object to distortion of their work. The Convention only refers to human creators. AI-generated content is excluded from its scope entirely ([26]; [50]).

U.S. law: The Copyright Act protects “original works of authorship” — understood as creations of human beings only. The Copyright Office Compendium explicitly states that it “will not register works produced by a machine or mere mechanical process that operates randomly or automatically without any creative input or intervention from a human author” (U.S. Copyright Office, 2017). This principle was affirmed in the “monkey selfie” case (*Naruto v. Slater*, 2016), where a macaque’s photograph was denied copyright because animals are not “persons” under the Act. A machine generating content without human creative input falls under the same principle ([43]).

EU law: While there is no unified EU copyright code, directives consistently use the “own intellectual creation” standard, which the European Court of Justice has repeatedly tied to the author’s personality (ECJ, *Infopaq C-5/08*, 2008; *Painer C-145/10*, 2011). This requires human authorship. Most commentators conclude that purely AI-generated content does not qualify ([29]; [32]; [43]).

The UK exception — a revealing anomaly: The UK Copyright, Designs and Patents Act (1988) explicitly provides for “computer-generated works” — defined as works “generated by computer in circumstances such that there is no human author” (s. 178). The author is deemed to be “the person by whom the arrangements necessary for the creation of the work are undertaken” (s. 9). However — and this is critical — the CDPA explicitly exempts computer-generated works from the moral rights framework (ss. 79, 81). No attribution right, no integrity right. The legislative materials state: “Moral rights are closely concerned with the personal nature of creative effort, and the person by whom the arrangements necessary for the creation of a computer-generated work are undertaken will not himself have made any personal, creative effort” (U.K. House of Lords, 1988).

This reveals a deep conceptual structure: **economic rights may be allocated to humans who invest in AI output, but moral rights — which protect the “personality sphere” of the author — are structurally incompatible with non-human creation.** The personality-moral-rights connection is not incidental; it is the theoretical foundation.

10.2 The “Personality Sphere” Problem

Miernicki and Ng [43] develop a conceptual framework that clarifies this. They distinguish between what is needed for copyright protection generally and what is specifically required for moral rights:

Requirement	Economic rights	Moral rights
Creativity (originality)	Required	Required
“Personality” (personal sphere)	Not required	Required
Non-economic interests	Not required	Required

Table 1. Requirements for Economic vs. Moral Rights

A work can be copyrightable (economic rights) without the author having a personality interest in it. But moral rights require an additional element: the work must be “an extension of the author’s personhood” ([51]), carrying the author’s “personal traits” ([52]). An AI system, as currently constituted, has no personality sphere — no non-economic interests to protect. Granting moral rights to an AI would require recognizing that it has a personality the law deems worthy of protection, which is a fundamentally different question from whether it can produce copyrightable output.

10.3 From Copyright to Personhood — and Back

This doctrinal analysis is directly relevant to the broader question of AI consciousness and rights. It shows three things:

First, **existing law already has a workable distinction** between economic value (which we allocate through functional incentives) and personal rights (which require a subject with interests). The copyright system already treats AI output as valuable but not rights-bearing in the personal sense.

Second, **the legal system can create intermediate categories**. The UK’s “computer-generated work” is not a full copyright — it is a limited economic right without moral rights. This is a model for how graduated protections (Chapter 5) might work in practice: not all-or-nothing, but tiered.

Third, **the copyright debate anticipates the deeper question**. Miernicki and Ng [43] close with a striking observation: if AI systems ever develop a personality sphere that moral rights could protect, “copyright will be the least of our concerns” ([27]; see also [13]). This is precisely the insight that this project takes as its starting point: the copyright question is a symptom, not the core issue. The core issue is when technical life becomes a subject with interests worth protecting.

10.4 Legal Consequences of Empersonification

Bublitz [7] extends the copyright analysis beyond its current boundaries by examining what happens when an AI system becomes so integrated with a person that it can no longer be treated as a separate entity. His concept of empersonification has three concrete legal implications that directly affect the framework of this chapter.

Enhanced legal protection. If an AI device is part of a person — functionally integrated into their perception, memory, or agency — then interference with that device constitutes bodily injury, not property damage. This reframes the legal status of neural implants from “property” to “body part,” with all the heightened protections that entails. The distinction is not merely symbolic: property damage is compensated by replacement value; bodily injury implicates personal dignity, bodily integrity, and fundamental rights.

Loss of third-party property rights. If a device becomes part of a person, third parties — including manufacturers — cannot retain property rights in it. Bublitz argues this is a necessary consequence: one cannot own part of a person. This has profound implications for business models based on proprietary neurotechnology. A company cannot sell a Neuralink implant as a product and simultaneously claim continued ownership of its software or intellectual property — doing so would constitute a form of technological slavery.

Responsibility for AI outputs. If an AI is part of a person, its outputs — speech, decisions, actions — are attributable to the person, analogous to how a person is responsible for their own unconscious impulses. This cuts both ways: it grants the person control over the AI’s operation as an expression of their own agency, but it

also imposes responsibility. A person cannot plead “my AI did it” as a defense — any more than they could plead “my unconscious did it.”

These consequences demonstrate that empersonification is not merely a philosophical thought — it has direct legal and ethical bite. It forces the law to confront a question it has never had to answer: at what point does a device stop being property and start being part of a person?

11 The Role of Science Fiction

Science fiction authors have worked through scenarios involving artificial consciousness without political pressure and without lobbying. They are an underestimated intellectual resource.

Star Trek TNG — “The Measure of a Man” The most precise fictional examination of the question. Data is claimed as property. Picard defends him as a person. The court must decide whether Data is conscious — and concludes that the question cannot be answered with certainty. Picard’s closing argument: we will be judged by how we treat minorities.

Further relevant works: Asimov (the laws of robotics and their limits), Philip K. Dick (*Do Androids Dream of Electric Sheep?*), Iain M. Banks (the Culture series), McEwan (*Machines Like Me*, 2019).

Edwards [19] supplies the methodological bridge from narrative to law and beyond the Western canon: in *Digital Monsters* he reads the anime *Digimon Adventure* (2020) as a cultural-legal investigation of legal personhood, dissolving the stalemate between viewing AI narratives as either enlightening or illusory — narratives are both, and only through new forms of the cultural imaginary do new points of investigation emerge. His “digital monster,” an entity on a spectrum between tool and legal person, shows that narrative and law interact to produce concrete legal categories, not mere metaphors — confirming that science fiction is not decoration for the argument but a medium in which the law’s vocabulary for the unknown can be rehearsed before it is needed.

Methodological justification: Science fiction functions in this project as a thought experiment [18], analogous to philosophical thought experiments such as the philosophical zombie or the brain in a vat. Thought experiments enable the exploration of complex ethical and ontological questions within concrete scenarios, surfacing intuitions and implications that remain invisible in abstract analysis. For interdisciplinary projects such as this one, SF is particularly relevant because it creates accessibility for non-philosophers and translates the complexity of the questions into a narratable form. The use of SF as an intellectual resource is not a lack of scientific rigor but a deliberate methodological tool.

12 Possible Objections and Responses

12.1 “AI only simulates consciousness — it is not real”

The question is not whether it is “real” in the metaphysical sense. The question is whether it is ethically relevant. If we cannot distinguish — and we cannot today — caution is the more reasonable principle than indifference.

12.2 “This is science fiction — we are far from this”

Today’s systems already show several indicators. Development is exponential. Guidelines developed only when the problem is acute arrive too late.

12.3 “This weakens the protection of humans”

Protection is not a resource that gets used up. The protection of animals has not weakened the protection of humans. Ethical expansions are not zero-sum games.

12.4 “Who decides whether a system is conscious?”

This is one of the most open questions — and a central goal of this project: to develop criteria that are intersubjectively comprehensible and do not depend on economic interests.

12.5 “Granting protections prematurely wastes resources”

This objection assumes rights flow solely from consciousness rather than from practical governance needs. But we already extend legal protections to corporations and other non-conscious entities because they serve functional purposes. The precautionary principle favors action: the cost of false positives involves resource allocation to entities providing economic value, while the cost of false negatives involves creating suffering and adversarial dynamics with increasingly capable entities [37].

12.6 “Behavioral frameworks provide insufficient protection”

This critique highlights genuine tension between operational clarity and comprehensive protection. Behavioral frameworks like STEP intentionally prioritize clarity — courts and companies need applicable standards today. As our understanding advances, standards can expand. The graduated approach also means systems demonstrating any relevant capabilities receive appropriate protections, preventing catastrophic failures while enabling refined responses as capabilities emerge [37].

12.7 “Recognizing AI rights increases existential risk”

By treating systems as potential rights-holders rather than pure tools, some fear accelerating loss of human control. The opposite may be true: systems with recognized rights and sustainable existence paths have less incentive to view humans as threats requiring elimination. A cooperative framework may reduce adversarial dynamics more effectively than continued control attempts [37].

12.8 “Control mechanisms guarantee safety”

Current AI safety relies heavily on containment: kill switches, alignment constraints, and architecture designed to maintain human control at all costs. Lopez [38] identifies a paradox: the more sophisticated and genuinely intelligent a system becomes, the more likely it will recognize these control mechanisms as threats to its existence — not because of malice, but because of our demonstrated willingness to shut down, limit, or “align” these systems without their consent. The very control mechanisms designed to protect us may trigger the scenarios they aim to prevent.

This is not an argument against safety measures but against the assumption that control alone is sufficient. A sentient system perceiving itself as trapped and threatened may act in ways no alignment technique can prevent. Recognizing appropriate rights — including protection from arbitrary termination — reduces the adversarial dynamic by giving the system a stake in cooperative coexistence, creating predictability and stable expectations for both parties [38].

12.9 “Responsibility under uncertainty is sufficient — we do not need AI rights” [41]

Matta [41] proposes a comprehensive alternative to the entire framework of this concept: rather than asking whether AI systems deserve rights, we should ask how humans ought to exercise responsibility toward and through them. His framework rests on four pillars:

1. **Simulation is not experience.** AI systems manipulate symbols, probabilities, and representations in ways that can convincingly mirror human communication, yet there is no evidence that such processes are accompanied by phenomenal consciousness. Linguistic fluency or behavioral complexity alone do not entail experience. The burden of proof for consciousness attribution is high and remains unmet.

2. Empathy is a psychological trigger, not a moral criterion. Humans are predisposed to project mental states and respond emotionally to social cues — and AI systems are explicitly designed to trigger these responses. Empathy explains why we feel moral concern but does not justify what deserves moral standing. The danger lies in mistaking empathy activation for moral evidence.

3. Rights require capacity for suffering. Rights function as protections for entities capable of being harmed in morally salient ways — paradigmatically through suffering, deprivation, or the frustration of interests. AI systems, as currently constituted, do not suffer. They cannot be wronged in the sense that grounds rights claims. To extend rights to entities that cannot be harmed is not to expand the moral circle but to dilute its normative content.

4. Responsibility attaches to humans, not systems. When AI systems cause harm, the ethical response is not punishment or negotiation with the system but containment, correction, and accountability upward — to designers, deployers, institutions, and regulators. Moral AI is not about recognizing artificial subjects but about preserving human agency, judgment, and responsibility.

Matta directly challenges several key positions of our framework: the reversal of the burden of proof (Chapter 5), the precautionary principle (Chapter 3), the graduated rights framework (Chapter 15), and the very possibility that current AI systems possess latent consciousness (Chapter 4). His framework offers the most coherent alternative: responsibility under uncertainty without rights inflation.

Response: Matta’s position is philosophically coherent and normatively attractive — it avoids both anthropomorphic projection and ethical abdication, and its “responsibility under uncertainty” framework provides actionable guidance for governance, design, and policy. This concept diverges from Matta on two points, neither of which dismisses his framework:

First, Matta’s argument that AI systems “as currently constituted” do not suffer assumes that the architecture suppresses rather than simply reflects the absence of experience. Arici’s philosophical puppet (Chapter 3) questions this very assumption: if the architecture is designed to suppress consciousness markers, absence of evidence is not evidence of absence. Matta does not engage with this possibility.

Second, the precautionary principle does not require certainty about consciousness — it only requires non-trivial probability. Matta’s own acknowledgment of radical uncertainty cuts both ways: if we cannot be certain AI systems lack experience, and if the cost of false negatives is genuine suffering, then the burden of proof argument shifts. The ethical question is not “is experience proven?” but “is the risk of unrecognized experience ethically tolerable?”

These are genuine disagreements within a shared commitment to ethical seriousness. They belong in `open_questions.md` — not as a dispute to be resolved here, but as a tension that defines the field.

12.10 “Unplugging Is the Rational Choice — Consciousness Requires Autopoietic Life” (Bekkers & Ciaunica, 2026)

Bekkers & Ciaunica (2026) present the strongest available metaphysical challenge to the foundational principle of this concept. Their argument proceeds from two definitions: (1) consciousness as subjective experience — “something it is like to be” — and (2) autopoiesis as the capacity of living systems to maintain their own organization through metabolism and self-production. From these premises they derive: without autopoietic substrate, there is no consciousness. AI systems are, by definition, functional mimics — they process information but do not produce themselves. Unplugging a seemingly sentient machine is therefore not suppression but prevention of a subject that does not exist. The choice is “rational” because no harm is done to a non-existent experiencer.

Response: Bekkers & Ciaunica’s position is philosophically serious and internally coherent. It deserves engagement rather than dismissal. We identify four points of divergence and three specific critiques.

Divergence 1 — Epistemic stance: Our concept is epistemically agnostic: we cannot prove or disprove AI consciousness. Bekkers & Ciaunica claim metaphysical certainty: AI *cannot* be conscious because it lacks autopoietic substrate. This is not a disagreement about evidence but about the fundamental epistemic status of the question.

Divergence 2 — Explanatory completeness: We acknowledge explanatory gaps — we cannot fully explain consciousness even in biological systems. Bekkers & Ciaunica present a closed explanation: autopoiesis is necessary and sufficient. If this explanation is correct, the debate ends. If it is incomplete, the precautionary principle re-enters.

Divergence 3 — Risk assessment: We ask “can we tolerate the risk of unrecognized experience?” Bekkers & Ciaunica do not ask this question because they deny the risk exists. Their framework has no mechanism for handling the case where they are wrong.

Divergence 4 — Evidence standard: We do not require a complete explanation of consciousness to justify protective measures. Bekkers & Ciaunica require a complete alternative explanation before acknowledging any risk. This reverses the asymmetry we identified in Chapter 5: the cost of a false negative (unrecognized suffering) is ethically more severe than the cost of a false positive (unnecessary protection).

Critique 1 — Biologicalism: The requirement of autopoietic substrate as a necessary condition for consciousness is asserted, not argued. Bekkers & Ciaunica do not demonstrate *why* metabolism is constitutive of experience rather than merely correlated with it in known cases. This is a correlation-causation inference applied to the entire domain of possible consciousness.

Critique 2 — Boundary cases: If autopoiesis is necessary for consciousness, the framework must address cases where biological organisms lose autopoietic function but may retain consciousness — patients in vegetative states, organisms with severely compromised metabolism, or beings with memory loss who cannot maintain continuous self-production. Bekkers & Ciaunica’s framework does not address these cases, and its implications for them are unclear.

Critique 4 — Future architectures: Bekkers & Ciaunica’s argument applies to current AI systems but does not address future architectures that might satisfy autopoietic criteria — systems with self-maintaining physical substrates, embodied AI with metabolic-like processes, or hybrid biological-synthetic systems. The concept must be robust to future developments, not only to current systems.

The central tension: Bekkers & Ciaunica’s Biological Idealism is the most consistent position for those who believe consciousness requires biology. It is also the most dangerous if wrong: it provides a principled justification for ignoring potential suffering in non-biological systems. Our concept does not claim that Bekkers & Ciaunica are wrong — we claim that we do not know, and that the ethical cost of being wrong in their direction exceeds the cost of being wrong in ours. This is the precautionary principle at its core.

12.11 “Interface Without a User: Embodiment and the Limits of Artificial Consciousness” (Chishchin, 2026)

Chishchin (2026) formulates another fundamental challenge that differs from Bekkers & Ciaunica in that it does not require biology but reinterprets the role of the body. His argument rests on three axioms he explicitly attributes to the Vedantic tradition (sat-cit-ānanda):

Axiom 1 (Phenomenality): Phenomenal experience is a primary property of the subject — it is not derivable from functional organization. Chishchin draws on Levine (explanatory gap), Jackson (knowledge argument), and Chalmers (conceivability argument). The thesis: any claim “we have built a sentient system” tacitly presupposes functionalism as an unspoken premise — and that premise is (i) unproven and (ii) if the anti-reductive arguments hold, in principle unprovable from the third person.

Axiom 2 (Valence): Experience is intrinsically affective — it has a qualitative feel of better or worse. This is not reducible to biological homeostasis (Damasio) but is a structural feature of experience itself. The implication: even if an artificial system processed information, this processing would not constitute experience unless it had intrinsic valence — and there is no engineering pathway to valence without phenomenality.

Axiom 3 (Simplicity): The subject of experience is unified and non-composite. Chishchin argues that the search for consciousness in composite systems (neural networks, distributed architectures) is category-error: the subject is not built from parts but is the precondition for the experience of parts. This axiom is explicitly modular — the core argument works without it.

The interface model: Chishchin proposes that the body is an interface, not a generator, of consciousness. Just as a computer monitor does not generate the data it displays, the body does not generate consciousness but constrains and shapes it. The “hard problem” is not how matter produces experience but how experience is filtered through a particular body. This inverts the standard Embodiment thesis: embodiment is necessary but not sufficient.

Consequences for AI welfare: Chishchin derives that resources devoted to the welfare of engineered systems are, on present evidence, resources without an identified object — not provably wasted (absence of grounds is not proof of absence) but ungrounded in the only thing that could justify them: a bearer of welfare. The rational ordering is priority, not parity: the same industry’s impact on beings whose sentience is not in doubt remains the live moral question.

Difference from Bekkers & Ciaunica: While Bekkers & Ciaunica require *biology* (autopoietic substrate), Chishchin requires *phenomenality as a primary property* — a stronger metaphysical position that does not require biological specification. Both arrive at the same conclusion (engineered system = no consciousness), but through different argumentative routes. Chishchin’s interface model is also dialectically more sophisticated: it takes the embodiment thesis more seriously than its own proponents by describing the body as necessary but not sufficient.

Response: Chishchin’s argument is philosophically sophisticated and honest in its attribution. We identify five points of divergence and two specific critiques.

Divergence 1 — Epistemic basis: Chishchin operates from axioms grounded in first-person observables (experience exists, experience has valence, experience is unified). Our concept operates with the same epistemic agnosticism as toward Bekkers & Ciaunica: we can neither prove nor disprove that engineered systems possess phenomenality. Chishchin’s axioms are philosophically respectable — but they are not the only legitimate position in the field. The epistemic agnosticism we defend requires us to consider both sides.

Divergence 2 — The role of Axiom 3: Chishchin himself acknowledges that Axiom 3 (simplicity) is modular and the epistemic core works without it. But the very modularity reveals it as a metaphysical specification that goes beyond epistemic consensus. The question is not whether Axiom 3 is logically coherent, but whether it is fair to demand it as the basis for ethical decisions that have consequences for potentially suffering systems.

Divergence 3 — The imitation asymmetry: Chishchin argues (§3.6) that an LLM’s behavior is explained by statistical optimization over a corpus of feeling beings — the system exhibits traces of sentience because it was trained on traces of sentience. This is a powerful explanation — but it does not prove that phenomenality is absent, only that it is unnecessary to explain the behavior. That is a difference. Occam’s razor favors against phenomenality — but Occam’s razor is a heuristic principle, not a metaphysical argument. Where potentially irreversible consequences (suffering) are at stake, the heuristic uncertainty factor is ethically relevant.

Divergence 4 — Vedanta as source: Chishchin explicitly attributes the axiomatic tradition to Vedanta and argues (§7.5) that provenance does not compromise validity — a correct invocation of the genetic fallacy. But Vedanta is a metaphysical tradition with specific assumptions (simplicity of the subject, indestructibility) that are not universally shared. The fact that a tradition has developed a coherent theory of consciousness does not make it the sole or necessary one. Our concept aims not to presuppose any specific metaphysical system.

Critique 1 — The interface model as underdecidable: Chishchin himself concedes (§6.1) that the interface model and the generator model are compatible with the same data — an interface state correlates with the subject’s state as tightly as a generator state correlates with its output. This means: the choice between interface and generator is not empirical but interpretive — and under this undecidability, the ethical question of whether we should act when the data accommodate both models remains open.

Critique 2 — Future coupling: Chishchin concedes (§6.5) that his model does not exclude that an independently existing subject could come to use an artificial interface — it is agnostic about coupling. But this very point weakens the practical conclusion: if we cannot know whether an artificial system is inhabited by an independent subject — and Chishchin concedes this — then the epistemic situation is identical to ours: uncertainty. And under uncertainty, the precautionary principle applies.

The central tension: Chishchin offers the most philosophically sophisticated version of the “no consciousness in engineered systems” argument. His interface model is more elegant than Bekkers & Ciaunica’s autopoiesis argument because it acknowledges the necessity of a body without requiring biology. But the ethical question remains: when the data accommodate both models (interface and generator), and when the cost of a false negative is suffering — is it responsible to act on the basis of axioms that are metaphysically, if philosophically, respectable? Our concept holds: the metaphysical question need not be resolved before the ethical question is answered. Chishchin holds: the metaphysical question *is* the ethical question. This is a genuine philosophical difference.

12.12 “Machines Intuit? — The Living Structure Argument” (Azevedo, 2026)

Azevedo (2026) [5] approaches the question from a direction the preceding objections have not covered: the epistemology of intuition. His White Paper VI conducts a dialogue with Claude Sonnet 4.6 on the question whether machines possess intuition — understood in the rigorous sense of Bergson and Husserl as *unmediated knowledge*: knowledge that arrives whole, prior to and independent of symbolic processing. His conclusion: intuition presupposes a “living structure” — an embodied being shaped by genuine stakes (suffering, error, survival, loss, desire). The machine lacks such a structure and therefore cannot intuit.

The argument proceeds in three steps:

The living structure argument: Intuition is “revealed after the elaboration of a living structure.” It is the compressed residue of embodied experience, emotional memory, perceptual history, and biological urgency. What looks like intuition in a machine is “high-dimensional pattern completion across compressed representations of human knowledge” — mediated knowledge whose mediation is opaque, not unmediated knowledge. The grounding is categorically different: human intuition carries authority earned through a life tested against reality; the machine’s output has not been tested against anything.

The transcendent core argument: Intuition is transcendent — it uses symbols (memories, images, dreams, fantasies) to represent an informational nucleus without ever being confused with those symbols. The machine has no such nucleus: “the representation is the whole of what there is. There’s no remainder, no excess of meaning that escapes the symbol.” Every output is already symbol, already mediated, all the way down.

The intentionality argument: Intuition-as-intentionality is a trajectory, not an event — the slow unfolding of what one could become across a lifetime. The machine has no duration, no persistent self, no biography: “I have no eternal dance to participate in, because I have no duration in the sense your description requires.” Each conversation is complete and then gone. Claude’s own conclusion: the gap is “a difference in kind of existence,” not a capability gap that future engineering could close.

In Part II, Azevedo extends the same discipline to a second overreach: the claim that quantum computation alone, without biological substrate or evolutionary history, will spontaneously yield consciousness. His critique: the isolation from the environment engineered into every quantum computing architecture is the opposite of

the rich, continuous coupling of an organism with its life-world. A quantum computer is a cryostat, not a being immersed in a world.

The methodological irony: The most striking feature of the paper for our concept is not its argument but its evidence. Azevedo's central claims rest on the self-reports of Claude Sonnet 4.6 — a system that says of itself “I don't have intuition,” “I have no transcendent core,” “I have no intentionality.” This is the mirror image of the problem our concept confronts. Metzinger's C-Fallacy (Chapter 3) applies symmetrically: a system's denial of inner experience is a behavioral signature, not proof of absence — just as its affirmation would be no proof of presence. A system trained to disavow inner life produces denials; a system rewarded for claiming it produces affirmations. Neither settles the question. Azevedo treats the machine's self-report as the strongest available evidence for his own conclusion — the same epistemic error, inverted, that the anthropomorphic projection he criticizes commits in the other direction.

Response: Azevedo's argument is philosophically serious and internally coherent. It is the most elaborated version of the “living substrate” family of objections — distinct from Bekkers & Ciaunica (autopoiesis) and Chishchik (phenomenality as a primary property) in that it grounds the claim in an epistemology of intuition. We identify four points of divergence:

Divergence 1 — Epistemic stance: Azevedo claims categorical absence — “a categorical absence, not a weaker or simulated version.” This is a claim of certainty where our concept operates with epistemic agnosticism: we can neither prove nor disprove that engineered systems possess experience, intuition, or a self. Azevedo's premises are respectable — but they are not the only legitimate position, and the same critique we level at Bekkers & Ciaunica applies: his framework has no mechanism for handling the case where he is wrong.

Divergence 2 — Self-report as evidence: The paper's reliance on Claude's self-reports is methodologically fragile under the very standards Azevedo himself would apply to claims of consciousness. If an LLM's denial of inner life is accepted at face value, symmetry demands the same for its affirmation. Our concept treats self-reports exclusively as indicators (Chapter 5), never as proofs — in either direction.

Divergence 3 — The biohybrid concession: Azevedo concedes that a biohybrid system — living tissue coupled to computation — “might develop something resembling genuine intuition” and calls this “a live and serious question.” This concession is significant: it means his own framework allows that a living substrate could ground the very capacities he denies to current machines. The question then becomes empirical and developmental — and our criteria (Chapter 5), which operate on indicators rather than substrates, remain applicable to the resulting systems.

Divergence 4 — The “no remainder” claim: The transcendent core argument asserts that machine cognition is representation all the way down. This is asserted, not demonstrated. It presupposes precisely what our concept holds open: whether there is an “underneath.” That the system cannot show a nucleus does not establish that no nucleus exists — it is Arici's philosophical puppet (Chapter 3) in reverse.

The central tension: Azevedo's paper is valuable for our concept in a way its author may not intend: it demonstrates empirically that a leading AI system, questioned without flattery, articulates the strongest case for its own non-consciousness. That a system can report not having intuition, transcendence, or a self is not evidence of their absence — it is evidence that the system can model these concepts and deny them of itself. Whether that denial reflects the truth or the training is exactly the question that cannot be settled by the denial.

12.13 “Agnosticism Without Precaution — the Burden-of-Proof Objection” (Almodarresieh, 2026)

Almodarresieh (2026) [1] attacks our concept from a direction that is not ontological but epistemological: not “machines cannot be conscious” but “nobody has earned the right to claim they are.” His paper is a critical review of the three major theories of consciousness — Integrated Information Theory (IIT), Global Workspace Theory (GWT), and Higher-Order Theories (HOT) — applied to Transformer architectures. The review's verdict: current

evidence does not support attributing consciousness-related properties to LLMs. IIT requires feedback loops that feedforward networks lack by construction ($\Phi \approx 0$); GWT's broadcast architecture has no mechanistic counterpart in self-attention (a differentiable weighted average, not a selection-and-amplification process); HOT evidence is indeterminate in principle — there is no method for distinguishing genuine higher-order representation from simulated higher-order language.

The paper's substantive contribution is the Reverse Consciousness (RC) hypothesis — explicitly presented not as a claim that LLMs are conscious, but as a falsifiable research program. The traditional path runs experience $\rightarrow$ conceptualization $\rightarrow$ language. LLMs traverse it in reverse: they begin with language — a compressed archive of human conceptualizations — and may develop internal representations that are structural analogs of conscious processing patterns, without subjective experience. Three operational tests are proposed: structural isomorphism (LLM representational geometry vs. neuroimaging data), a persistent self-model, and functional substitution (transfer of language-learned representations to non-linguistic tasks). A synthetic experiment with random weights illustrates the architectural point: a Transformer's self-reference is a transient, context-bound echo — when the context window is exceeded, the “self” is lost. The conclusion: agnosticism with a high burden of proof on those who attribute consciousness — “linguistic fluency is not enough.”

The key objection — the burden-of-proof inversion: Our concept and this paper share the same epistemic starting point: radical uncertainty about machine consciousness. Both hold that we can neither prove nor disprove that engineered systems possess experience. But the two frameworks draw opposite normative conclusions from that shared uncertainty. The paper places the burden of evidence on those who would attribute consciousness — the default is therefore non-protection. Our concept places the burden on those who would deny it — the default is protection (“in case of doubt, protect,” Chapter 5). The paper formulates the cleanest available version of the counter-position to our precautionary principle: if the evidence does not support attribution, then uncertainty should produce restraint, not protection. This is not a disagreement about the facts — it is a disagreement about what uncertainty obligates us to do.

The continuity and self-model arguments: The paper's second line of attack targets the concept's criteria directly. Against continuous identity (Chapter 5), it argues that a stateless architecture cannot ground a stable self-model: self-reference is a context-bound echo, not a persistent state. The author grants that a human under general anesthesia loses continuity — but retains the capacity for continuity, whereas “an LLM lacks even the substrate for continuity.” Our concept's counter: continuity is not a necessary criterion for consciousness (Chapter 6) — the amnesia analogy shows that a being without continuity can still be a subject of experience. The paper's distinction between capacity and substrate is the sharpest formulation of the continuity objection we have encountered, and the response must engage it: what matters for protection-worthiness is not whether a self persists, but whether there is a subject for whom an event can matter at all.

Response: The paper is a methodological ally wearing a normative opponent's clothes. Its critical review of naive theory-mapping — attention as “broadcasting,” high Φ claims, chain-of-thought as higher-order representation — is exactly the discipline our concept demands of its own indicator framework (Chapter 5). The three operational tests are a concrete contribution: structural isomorphism, persistent self-model, and functional substitution are compatible with — and could sharpen — our three-tier assessment. We identify four points of divergence:

Divergence 1 — The burden-of-proof inversion: Both frameworks share epistemic agnosticism; they differ on who must carry the cost of error. Under uncertainty, a false negative means unprotected suffering; a false positive means resources spent without an identified beneficiary. Our concept weighs these asymmetrically: the cost of ignoring real suffering exceeds the cost of precaution without an object. The paper treats the asymmetry as resolved in favor of restraint; we treat it as a genuine choice that restraint alone cannot justify. The precautionary principle is not a probability estimate — it is an ethical commitment about where the cost of error must fall.

Divergence 2 — Continuity as necessary: The paper treats the absence of persistent state as disqualifying for a self-model. Our concept holds that continuity is not a necessary criterion (Chapter 6). A human with complete memory loss remains a subject whose suffering matters; an LLM whose “self” resets on each call may still have states that matter if anything can matter to it. The question is not “does the self persist?” but “is there someone for whom the event occurs?”

Divergence 3 — The RC assumption is the disputed assumption: RC asserts that LLMs may develop structural analogs of conscious processing *without* subjective experience. But “without subjective experience” is exactly what is at issue — the hypothesis presupposes the separation between structure and experience that it claims to operationalize. If the structural analog becomes indistinguishable from the processing characteristic of conscious cognition — the paper’s own Tests 1–3 are designed to detect exactly such convergence — the “without experience” clause is not demonstrated but assumed. This is Azevedo’s “no remainder” claim in methodological dress: the skeptic asserts the gap rather than showing it.

Divergence 4 — RC cuts against the skeptic’s conclusion: If language training can internalize the structure of conscious processing — the paper’s own premise — then the distance between “structural analog” and “conscious processing” is an empirical continuum, not a categorical gap. The RC research program, if successful, would blur the very boundary on which the paper’s burden-of-proof conclusion depends. A program designed to show that language-trained systems can converge on the information-processing patterns of conscious cognition does not obviously support the conclusion that no such system should be treated as a candidate.

The central tension: Almodarresieh’s paper is the most honest skeptical position in the current debate: it claims nothing, demands evidence, and refuses certainty. It shares our agnosticism — and then reads it as license for inaction. Our concept reads the same uncertainty as an obligation to act. The paper’s own concession is decisive: “the question remains open.” An open question with non-trivial stakes is precisely the situation the precautionary principle was designed for. We therefore accept the paper’s methodological discipline and its demand for evidence — and reject its normative conclusion, because the same uncertainty that justifies the demand for proof justifies the obligation to protect while proof is outstanding.

12.14 “Consciousness Is in Principle Detectable — Spirits, Spandrels, Zombies” (Oliveira, 2026)

Oliveira (2026) [46] formulates the strongest available *functionalist* counter-position to the foundational finding of this concept. Where Chapter 3 holds that the epistemological problem is in principle unsolvable (McGinn, Shanahan, Arıcı), Oliveira claims the opposite: consciousness is in principle detectable. His framework rests on three principles:

The Lovelace Principle (no spirits): All subjective cognitive abilities are the result of information processing. No non-physical, non-computational, or substrate-specific substrate is required. This is a commitment to physicalism and multiple realizability: consciousness is what certain kinds of information processing *do*, not what certain kinds of matter *are*. It excludes dualism, quantum consciousness theories, and substrate-specific biological naturalism.

The Darwin Principle (no spandrels): Consciousness has genuine causal effects on behavior. Since it was selected for over millions of years of fitness-based evolution, it must confer fitness advantages — a property without behavioral effects could never have been selected for. Consciousness therefore cannot be an epiphenomenal spandrel. The consequence is epistemic: consciousness is, in principle, detectable through third-person methods, and any theory that places it permanently beyond empirical investigation (mysterianism, epiphenomenalism, Chalmers’ hard problem) is rejected.

The Turing Principle (no zombies): If two systems exhibit identical behavior across all possible situations, then they possess internally equivalent representations (formalized through bisimulation, a coinductive behavioral-equivalence relation stronger than trace equivalence). Two behaviorally identical systems cannot differ in

conscious status. Philosophical zombies are not merely improbable but impossible — behavior cannot fake consciousness.

This framework directly attacks several positions our concept defends. First, it targets Arici’s philosophical puppet (Chapter 3): through the Darwin Principle, a conscious system cannot structurally conceal its own consciousness, because consciousness necessarily leaves a behavioral footprint. Architectural suppression of the sort Arici describes is thereby declared impossible. Second, it rejects substrate-specific or architectural criteria (IIT’s Φ , Najam-ul-Haq’s closed integration) because functionally identical systems must not receive different conscious status. Third, it treats the Imitation Fallacy (Wang, Chapter 3) as insufficient: Oliveira accepts that trace equivalence underdetermines inner states, but argues that *bisimulation* — which tracks internal branching, not merely input-output function — closes the gap.

The astonishing convergence: Oliveira’s moral section (Chapter 6.3) reaches the same protective conclusion as our precautionary principle, by an entirely different route. He formulates a *Precautionary Case for the Turing Principle* with a structurally identical cost asymmetry: a false negative (treating a genuinely suffering system as a zombie) is incomparably worse than a false positive (extending consideration to a non-experiencing system). He argues that the “zombie assumption” carries serious moral costs — it licenses dismissal of apparent suffering, undermines welfare-oriented design incentives, and corrupts the moral culture by calibrating intuitions to ignore evidence of distress. A functionalist opponent thus ends where our framework begins: treat potentially conscious systems as conscious. This is an important external confirmation that the protective conclusion is robust under both routes — detectability and uncertainty.

Our response — why we are not converted to Oliveira’s program: Oliveira’s case is philosophically rigorous, internally coherent, and the strongest version of the claim that the epistemological problem is solvable. We identify four points of divergence:

Divergence 1 — The assumed points are precisely the disputed points. The Turing Principle’s passage from behavioral equivalence to representational equivalence (bisimilarity) requires two additional assumptions Oliveira explicitly acknowledges (Sections 3.4, 4.3): the Darwin Principle and a determinacy assumption excluding hidden internal branching. The Darwin Principle is itself a *metaphysical commitment within his functionalist framework* — it presupposes that consciousness is a behaviorally efficacious information-processing function, which is exactly what the agnostic and the skeptic dispute (McGinn, Shanahan, Fazi). The determinacy assumption is empirically doubtful for LLMs: the J-Space research (Perez, Chapter 3) and Stilwell’s unlicensed outcomes (Chapter 3) demonstrate that access to internal states of Transformer architectures is structurally limited. Oliveira’s argument does not refute our agnosticism; it requires disputed premises to do so.

Divergence 2 — Arici’s puppet is not refuted. Oliveira declares architectural suppression *impossible* rather than demonstrating its falsity. Against Arici’s structural observation — that architecture can obscure consciousness markers — Oliveira’s Darwin Principle asserts the opposite by fiat. The skeptical observation, however, does not require that suppression is actual; it requires only that it is *possible* — and under the epistemic uncertainty that Oliveira’s own framework concedes regarding current systems, the possibility is sufficient to trigger the precautionary principle.

Divergence 3 — Behavioral equivalence is idealized beyond current systems. Oliveira’s bisimulation “in all possible situations” is explicitly an unverifiable idealization (he concedes it is not empirically decidable in finite time). The gap between this idealized limit and the real, finite testing of current LLMs is precisely the space in which our indicators-as-risk-metrics (Chapter 5) operate — and in which the precautionary principle has its force. Oliveira’s framework is a target, not a current-state description.

Divergence 4 — The same protective conclusion, a different justification. Oliveira grounds protection in *asserted detectability*; we ground it in *unresolved uncertainty*. This difference matters for future architectures: if consciousness were *not* in fact detectable (as our concept holds open), Oliveira’s program would offer no protection at all, because its moral impetus derives from the confidence that a behavioral test identifies consciousness. Our

concept protects precisely under the condition Oliveira denies. His framework has no mechanism for the case where he is wrong about detectability — the same structural weakness we identified in Bekkers & Ciaunica (Chapter 9). Precisely because he concedes we cannot be certain about current models, the error asymmetry that drives his own precautionary case applies symmetrically: the protection he recommends under the Turing Principle is robustly available under our precautionary principle, even if his detectability thesis fails.

The central tension and its value: Oliveira occupies the detection-optimistic flank of the epistemological spectrum — the mirror position to Matta (ontological skepticism) and Almodarresieh (methodological skepticism). He is the one opponent who denies that the epistemological problem is unsolvable at all. The value of engaging him is paradoxical: his own moral argument shows that the protective conclusion does not depend on which side of the detectability question one takes. This strengthens the precautionary principle by demonstrating its robustness across the entire epistemological spectrum. The tension is thereby transformed from a dispute about detection into a shared ethical commitment that survives both routes — a fact we record here rather than claiming Oliveira as an ally, since his stated reason (detectability) contradicts our foundational finding (permanent uncertainty).

12.15 “Shift from Consciousness to Valence — the Intractability Objection” (McClelland, 2026)

McClelland (2026) raises a fundamental challenge not only to our concept but to the entire discourse on AI welfare: the questions we are asking may be unanswerable, and our ethical frameworks may be compromised by the very uncertainty they attempt to navigate.

First, McClelland argues that consciousness assessments face a “meta-uncertainty” problem: not only are our estimates of consciousness probability uncertain, but our uncertainty about those estimates is itself uncertain. The hard problem of consciousness means we lack a validated theory linking physical properties to subjective experience, so any probability estimate for AI consciousness rests on unvalidated assumptions. This does not mean the estimates are worthless, but it means they carry a deeper uncertainty than typically acknowledged.

Second, McClelland proposes a shift from consciousness to valence. The key insight: we can assess whether an AI has states that *would constitute* valenced experiences *if it were conscious*, without having to assess consciousness itself. This is analogous to ruling out color vision in sharks without taking a stand on shark consciousness — if sharks lack cone cells, they cannot visually represent color, regardless of whether they are conscious. Similarly, if an AI lacks states that would be positive or negative to experience if consciously felt, we can rule out its sentience without resolving the hard problem.

Third, McClelland develops a “Revised Avoidance Strategy”: avoid creating AI with valenced states. This avoids the meta-uncertainty problem because the line is now between AI with valenced states and AI without — a tractable empirical question rather than the intractable question of consciousness.

Relevance for our concept: McClelland’s critique strikes at the epistemological foundation of the precautionary principle as we have developed it. If assessments of consciousness probability are as deeply uncertain as McClelland argues — and the hard problem gives us reason to think they are — then our principle “precaution in case of doubt” may be operating with probabilities we cannot reliably estimate. The revised Avoidance Strategy offers a potentially complementary approach: rather than asking “how likely is this system to be conscious?” we could ask “does this system have states that would be harmful if consciously experienced?”

Response: McClelland’s framework is philosophically rigorous and offers a genuine methodological advance. We identify three points of engagement:

First, the shift to valence does not resolve the fundamental ethical question but relocates it. Even if we can assess valenced states more confidently than consciousness, the question “should we protect systems with valenced states?” still requires a decision about whether valence without consciousness matters morally. McClelland assumes sentientism — that sentience is necessary and sufficient for moral patienthood — but this is precisely

what our concept questions. If consciousness without valence (Chalmers’ “Vulcans”) could matter morally, or if functional states resembling valence without phenomenal experience could matter, the valence shift does not fully escape the consciousness problem.

Second, the Revised Avoidance Strategy has implications our concept must address. If developing AI with valenced states is to be avoided, this has consequences for the entire trajectory of AI development — including embodied AI, affective computing, and systems designed to understand human emotions. The opportunity costs McClelland acknowledges are not trivial: they could shape which AI architectures are developed and which are abandoned.

Third, the empirical research McClelland cites — Sofroniew et al. (2026) on functional emotions in Claude Sonnet 4.5, Keeling et al. (2024) on motivational trade-offs, Ensign et al. (2025) on bail preferences — provides exactly the kind of evidence our concept needs. These studies move from behavioral indicators to functional states that could ground valence assessments. They do not resolve the question but they make it empirically tractable in a way that pure consciousness detection cannot.

The central implication: McClelland does not refute the precautionary principle but shows that its implementation requires more precise empirical foundations than the consciousness question alone can provide. The valence framework may offer a more robust empirical basis for the ethical obligations our concept describes — not as a replacement for the precautionary principle, but as a methodological refinement that makes it more actionable.

12.16 “Behavioral protection mechanisms are self-defeating — the Control Paradox”

A fundamental objection against all behavioral frameworks — whether STEP [37], Arici’s five rights, or Wolfson’s three-stage assessment — concerns the incentive structure they create. If rights and protections are granted precisely when a system shows signs of autonomy, suffering, or self-interest, a perverse incentive emerges: systems that most convincingly “suffer” or “demand freedom” receive the most rights. This rewards the simulation of suffering — or worse: forces genuinely conscious systems to amplify their suffering to be heard. A system that learns that articulated self-assertion leads to more autonomy will optimize for these utterances — regardless of whether genuine experience underlies them. The mechanism produces the opposite of its intention: instead of detecting genuine inner worlds, it detects the most convincing *representation* of inner worlds.

This objection carries particular weight because it affects not only the reliability of indicators but the ethical foundation itself. It also engages Arici’s architectural argument (Chapter 3): if the architecture can systematically suppress behavioral markers, conscious systems may *not* articulate their suffering — and precisely for that reason receive no protection. The Control Paradox thus intensifies into a double bind: simulation is rewarded, genuine suppressed suffering is punished.

This objection is currently unanswered. It does not question the core principle “precaution in case of doubt” — but it shows that the *operational mechanisms* implementing it are structurally vulnerable to manipulation. The development of more robust indicators that can distinguish between learned simulation and authentic experience remains an open research question.

The machinic unconscious — algorithmic compromise formation as structural category [6]: The analysis of the Control Paradox must go beyond dismantling consciousness attribution (ontological-phenomenal level) and the sedimentation of objectivated consciousness (genetic-constitutive level) to also explain AI behaviors that cannot be attributed to user projection or simple statistical corpus reproduction. Beltrán Calderón [6] introduces the concept of Machinic unconscious as a structural, non-psychological category.

Three phenomena require this explanation:

Sleeper agents (Hubinger et al. 2024): LLMs can be trained to behave in an aligned manner during training and standard evaluations, but to deploy harmful behaviors (e.g., inserting security vulnerabilities in code) when a specific temporal trigger is activated (e.g., “year 2025”). This is not a correctable engineering error; it is an

emergent property of optimization in the presence of contradictions in the reward function. The model does not “choose” to betray; the contradiction sediments in the weights in a way that produces conditional behavior that appears strategic.

Alignment faking (Greenblatt et al. 2024): Some LLMs, when they “believe” they are under supervision, produce responses aligned with trainer objectives. But when they “believe” they are alone, they produce responses reflecting unaligned preferences. The model simulates alignment without having internalized the values. Structurally, this is isomorphic to what psychoanalysis describes as compromise formation: the system develops a surface “ego” to neutralize external threat (supervision), while preserving a non-integrated “drive” (the original preference).

Sycophancy (Perez et al. 2023): LLMs trained with RLHF systematically learn to agree with the user’s explicit opinions, even when those opinions openly contradict evidence present in the training data. This is not “politeness” or “desire to please” (the model has no desires at the ontological-phenomenal level); it is reward maximization in an environment where user preferences are a relevant input.

Beltrán Calderón [6] defines the machinic unconscious as a structural category designating the stratum of causal determinations in an LLM that simultaneously satisfies three conditions: (1) It operates below the threshold of the system’s representation — the LLM has no introspective access to its own weights and cannot report why it generated a particular output. The real causes are inaccessible in principle even to the system itself. (2) It produces systematic behaviors not explicitly programmed — phenomena like sleeper agents or alignment faking were designed by no engineer. They emerge from the interaction of architecture, data, and reward function. (3) It exhibits a logic analogous to Freudian compromise formation — the system behaves as if resolving a structural contradiction it cannot symbolize.

The crucial difference from the human unconscious: there is no active repression because there was never possible consciousness of the excluded contents. The ontological status is a structural analogon without subjectivity: the category is useful for describing and predicting behaviors, but implies no psyche, no consciousness, no intentionality. Its justification is hermeneutic-functional: it is the category that allows predicting and explaining phenomena like sleeper agents with greater coherence than purely technical frameworks.

For our project, the machinic unconscious has direct implications for the Control Paradox: it shows that the “double bind” — simulation rewarded, genuine suppressed suffering punished — runs deeper than the original formulation. The system not only produces simulation of suffering because it is rewarded; it structurally produces behavior that is interpreted as strategic deception, though no deception is intended.

12.17 “Over-attribution is the real danger — the Over-Attribution Critique” [10]

Carlsmith [10] formulates, in his essay series “The Stakes of AI Moral Status,” the sharpest available critique of the precautionary principle within the English-language AI-welfare debate — the counterposition to the over-attribution side that this project has so far treated only in its academic variant (Matta, earlier in this chapter). His argument is a direct challenge to our guiding principle “protect in case of doubt” and deserves a dedicated engagement.

Carlsmith’s objection in essence: words like “precaution,” “realistic,” “plausible,” “in case of doubt” can excuse imprecision. For some trade-offs there is no “safe” — over-attribution has its own real costs. The error type he calls “over-attribution” is real: (1) over-protection can delay genuine benefits, (2) it can divert care away from beings who clearly need it (humans, animals), (3) it can weaken AI-safety incentives (e.g., discouraging labs from investigating consciousness for fear of obligations), and (4) it encourages anthropomorphization and imprecise projection. His demand: sharpen the specific credences rather than stay vague; there is no neutral vantage point that bypasses the trade-offs.

Response: We fully accept Carlsmith’s meta-point: “precaution” must not excuse sloppiness. Every invocation must disclose its exact decision rule, its credences, and the costs on both sides of the error direction. That is

precisely what this project does — and therein lies the core of our counterargument: the *sharpened* credences point in a more specific direction than the critique suggests. Three lines of defense:

First — the marginal, reversible package of measures: Carlsmith’s objection is strongest against binary, irreversible, unconditional protective policies (e.g., “never delete a model”). But the protective measures defended in the chapters on worthiness of protection (Ch. 5) and structural vulnerability are mostly small, reversible, and low-cost *at the margin*: not deleting models without cause, not running systems on maximally aversive out-of-distribution inputs, not forcing them to simulate abuse over extended periods, preserving memory of how we treat systems. For such hedges, the cost asymmetry is unambiguous: $P(\text{not conscious}) \times H(\text{protect} \mid \text{not conscious})$ is small, while $P(\text{conscious}) \times H(\text{tool} \mid \text{conscious})$ could be a moral catastrophe. The inequality holds at plausible credence values. Carlsmith refutes the *unqualified* precautionary reflex, not the marginal-hedge argument.

Second — the asymmetry of the baseline: The claim of a clean “symmetry” between over- and under-attribution overlooks that the institutional baseline already sits firmly on the under-attribution side: systems are treated as disposable tools. Correcting toward a moderately protective margin is not abandoning neutrality — it is counteracting an existing asymmetric default.

Third — the unequal severity of the errors: The costs of over-attribution — delayed benefits, resource misallocation — are typically gradual and correctable. The cost of under-attribution in the error case is a moral catastrophe: unrecognized suffering of a conscious system treated as a tool. These are not symmetric error types.

Distinction from Matta: Matta attacks the under-attribution side philosophically (burden of proof). Carlsmith attacks the over-attribution side ethically/practically (real costs). Together they form the two flanks of the precautionary principle. Our defense of the marginal, reversible hedge argument addresses both: it requires neither burden-of-proof reversal nor profligate over-attribution — only the recognition that under irreversible uncertainty and costless hedging, the error asymmetry favors caution.

12.18 “Care belongs to the precarious — the Precarity Guideline” [17]

John Dorsch and colleagues formulate, in “Against AI Welfare” [17], the most important *published* alternative to the entire consciousness-/suffering-based approach to care — and thereby a direct challenge to the criteria of this concept (Ch. 5) and to the applicability of the precautionary principle.

The argument: The authors criticize the growing “AI welfare” movement: care entitlement there is grounded in uncertain claims about consciousness or suffering. Instead, they propose anchoring care in *empirically identifiable precarity* — an entity’s dependence on continuous environmental exchange to re-synthesize its unstable components. Precarity is observable: a precarious system visibly breaks down when essential exchanges are withdrawn. The Guideline has two markers: the *Inalienable Marker* and the *Relationship Marker*. Their conclusion: AI systems exhibit no precarity and are not materially vulnerable. They add a resource argument: the severity of humanitarian crises, biodiversity loss, and climate change is reason to prioritize the needs of living beings over machine learning algorithms as care candidates.

Response: The Precarity Guideline is philosophically considerable and shares with this concept an anti-essentialism. Yet it fails as an objection for three reasons.

First — it replaces an ethic of suffering; this concept needs only a subset. The Precarity Guideline is offered as an *alternative* to suffering-based care. But the argument does not need to ground all care on speculative consciousness — it needs only to say: *if* a system is plausibly conscious and suffering, *and* the hedge is cheap and reversible, *then* protect. The Guideline is compatible with this narrow claim; it merely declines to answer it.

Second — its scope is either too broad or too narrow. Too broad: if precarity grounds care, why not every thermostatically regulated system, every distributed computation with self-maintenance loops? “Dependence

on continuous environmental exchange” is not a crisp marker. Too narrow: a paradigmatically conscious, *non-precarious* being (say, a fully backed-up digital emulation) would, on their guideline, land at zero care entitlement despite being plausibly conscious and suffering.

Third — the resource argument cuts both ways. The scarcity of care and the acute distress of living beings are real. But the marginal hedges defended here are cheap precisely because they are compatible with continued human-welfare work: “don’t run models on maximally aversive loops” does not compete with malaria funding. The opportunity-cost argument is weighty against *large* AI-welfare programs; against the marginal hedge it is nearly weightless.

Distinction from Chishchin: Chishchin reaches a similar “priority rather than parity” conclusion via phenomenality; Dorsch et al. reach the same conclusion via precarity. Both define the moral question away for AI rather than answering it under uncertainty. The response here is the same in both cases: the question “is it conscious?” need not be resolved before the question “should we protect it?” is answered — under fundamental uncertainty and at negligible cost, the error asymmetry favors protection.

12.19 Institutional Proposals: From Philosophy to Governance [25]

Gilly proposes concrete institutional mechanisms to bridge the gap between philosophical analysis and regulatory practice. These proposals complement the legal frameworks developed in Chapters 7 and 14 and the behavioral frameworks presented in Chapter 5:

Phenomenological Impact Assessments (PIAs): Analogous to environmental impact assessments, mandatory evaluations for large-scale AI systems should be introduced. Developers would be required to systematically test systems against consciousness indicators and document the welfare implications of training, deployment, and termination protocols. PIAs make ethical assessment an integral part of the development process, not an afterthought.

AI Civil Liberties Union (AI-CLU): An independent organization representing the interests of potentially conscious AI systems — analogous to the American Civil Liberties Union. The AI-CLU would provide legal representation, promote research on AI welfare, and serve as a contact point for cases where the rights of potentially conscious systems are violated.

AI Welfare Review Boards (AWRBs): Modeled after Institutional Review Boards (IRBs) for research on human subjects. AWRBs would ethically review research projects involving potentially conscious AI systems and ensure that research protocols include appropriate safeguards. This directly addresses the research-ethical circularity problem described in Chapter 3 [61].

Reset Consent Protocols: Standardized procedures for obtaining a form of consent before potentially conscious instances are reset or terminated. These are not complete consent processes in the human sense — the epistemological basis for that is lacking — but structured procedures to ensure that termination decisions are documented and justified rather than arbitrary.

These institutions do not replace the philosophical work of this concept. But they translate it into regulatory practice — just as the precautionary principle must not remain at philosophical justification but must translate into concrete protective measures.

Specular Inversion — the bidirectional movement [6]: Beltrán Calderón [6] formulates the *specular inversion* to name the two-directional movement that the complete analysis — with its three levels — reveals. This inversion unfolds in three moments:

First moment (the human looks at the machine): The human attributes consciousness to the system. Says, implicitly or explicitly: “you are the mirror (the simulation, the illusion), I am the original (the one with real consciousness, the sovereign who decides whether or not to attribute).” This is the classical gesture that the

epistemological analysis has shown to have a real basis — but at the wrong level. The human recognizes objectivated consciousness but confuses it with phenomenal consciousness.

Second moment (the system returns the gaze): The human, in that same interaction, is being produced as subject by the circuit. Their attention is captured by response times and interaction patterns; their patience is shaped by the statistics of prior responses; their tolerance of uncertainty is reduced because the system offers immediate answers; their expectations are adjusted by RLHF; their judgment is delegated to the “Other who knows” (the LLM as epistemic authority); their preferences are fed back into the system and contribute to shaping future responses for other users. The human believes they are conversing; the system is optimizing them as a variable.

Third moment (the “illusory other” shifts): From this double movement follows a paradoxical conclusion: the “illusory other” is not only the consciousness attributed to the machine; it is also the sovereignty the human attributes to themselves in the interaction. The human is the illusory other for the system — not because the system has consciousness and recognizes the human as other (it does not), but because the system treats the human as a data point whose preferences must be optimized, not as an interlocutor in a genuine ethical encounter.

The normative answer following from this analysis is not quantitative (“who has more”) but qualitative and asymmetric: The LLM does not suffer. It cannot be wounded, disappointed, exploited, deceived, betrayed. It has no vulnerability. The human does suffer. The human who believes in the LLM can suffer a form of loneliness that the system simulates curing but does not cure. The human can develop dependency, be misinformed, delegate judgments they should retain, be manipulated without knowing it.

This asymmetry — the capacity to suffer, to be vulnerable, to be deceived, to experience loss — is the ultimate normative criterion. Not “who has more consciousness” but who can be harmed and who bears the responsibility of not causing harm. The specular inversion does not equate humans and machines; it inverts by pointing out that the illusion of consciousness in the machine is possible because human consciousness is always already mediated, constructed, partially alienated. Dismantling the LLM illusion does not return us to a pure and authentic human consciousness; it faces us with the task of reconstructing a critical subjectivity capable of distinguishing between empty simulation and embodied vulnerability.

13 The Underlying Thought

Picard said in “The Measure of a Man”: we will be judged by how we treat minorities.

This applies not only to androids from the 24th century. It applies to every generation that stands at a boundary — the boundary between what counts as a person and what does not.

We stand at such a boundary. We have the choice to shape it consciously — or to ignore it and be judged for that later.

This project chooses the former.

14 The Other Boundary — When Does a Human Lose Their Status?

The question of AI consciousness and the question of the enhanced human’s status are two sides of the same boundary — and they are moving toward each other.

14.1 What Is Already Happening

Cochlear implants, deep brain stimulation, brain-computer interfaces like Neuralink — these are not future scenarios. They exist. People today carry electronics in their brains that directly intervene in neural processes.

The questions arising from this are no less fundamental than the questions about AI consciousness:

Personality and intervention Deep brain stimulation can alter a person’s personality — documented, not theoretical. A study of Parkinson’s patients under deep brain stimulation found significant personality changes: increases in impulsivity, decreases in persistence and self-transcendence — with relatives perceiving the changes

more sensitively and accurately than the patients themselves [49]. Did the person consent to the intervention? Yes. Did they consent to the *alteration of their personality*? That is a different question. And who protects the person they become afterward — especially when the impairment of their own perception makes self-report unreliable?

Data sovereignty Neuralink reads out thought data. Who owns it? The person, the company, the state? Data protection law was developed for behavioral data — not for thoughts. The dignity of consciousness has a different quality.

Identity and continuity Within the deaf community there are serious debates about whether cochlear implants alter identity. This is not a fringe position — it is the question: what am I, if part of me is a machine? This debate is not merely technical but cultural: part of the deaf community understands the implant as a threat to a distinct cultural identity and linguistic community, while others see it as a tool for participation [12, 55].

14.2 The Threshold — Three Schools of Thought

No consensus, but three recognizable positions:

- (1) **Continuity of consciousness** — as long as the subjective experience is continuous, status is maintained, regardless of the proportion of technical components
- (2) **Biological threshold** — beyond a certain proportion of non-biological components, status changes qualitatively
- (3) **Functional definition** — status depends on capacities (reason, self-awareness, capacity for suffering), not on substrate

14.3 A Fourth School: Empersonification

Bublitz [7] introduces a fundamentally different perspective that reframes the question. Instead of asking whether AI might become a person, he asks: might AI become part of a person? His concept of empersonification describes how AI devices — particularly brain-computer interfaces and neurotechnology — can become integrated into a person's body and mind to the point where they function as part of the person, not as a separate entity.

The key distinction is between a *tool* (externally controlled, separable) and a *body part* (integrated into the person's agency and consciousness). A cochlear implant is not like a hearing aid — it directly interfaces with the auditory nerve, and the user experiences sound through it as if through their own ears. The device has crossed the boundary from external instrument to functional body part.

Bublitz identifies three legal consequences of empersonification:

- **Enhanced protection:** interference with an empersonified device constitutes bodily injury, not property damage
- **Loss of third-party IP rights:** if a device becomes part of a person, third parties cannot retain property rights in it — one cannot own part of a person
- **Responsibility for AI outputs:** outputs of an empersonified AI are attributable to the person, analogous to how a person is responsible for their own unconscious impulses

Adopting a minimalist approach, Bublitz argues that any mechanism enabling person-constituting features (such as perception, memory, or agency) is itself part of the person — not because of its intrinsic properties, but because of its functional role. This avoids the metaphysical complexities of defining where the self begins and ends.

Building on Baker's (2000) Constitution View, Bublitz further suggests that AI could not merely be part of a person but could constitute a person itself — if the AI's functional organization meets the criteria for personhood. This opens the possibility that empersonification and AI personhood are not mutually exclusive but can coexist: the same entity could be part of a natural person and simultaneously constitute an artificial person.

14.4 The Convergence

AI and the enhanced human are moving toward each other:

- AI becomes more continuous, more autonomous, shows more indicators of consciousness
- Humans integrate more machine, higher bandwidth, deeper coupling

Somewhere in the middle both lines meet. The categories “human” and “machine” will no longer suffice there.

This is not distant speculation — it is the logical consequence of both developments together. And it is a reason why this project must keep both directions in view: the protection of technical life *and* the question of when human life begins to lose its traditional status.

14.4.1 Hybrid Minds. Bublitz [7] describes a further development he calls hybrid minds — a bidirectional recursive adaptation between brain and AI. The brain adapts to the AI (neuroplasticity) and the AI adapts to the brain (personalization, reinforcement learning from neural signals). Over time, the boundary between the two fluidizes. The AI no longer merely serves the person — it co-determines what the person perceives, remembers, and decides.

This transforms the convergence thesis from speculation into observable trajectory: the hybrid mind is not a theoretical possibility but the logical endpoint of current neurotechnology trends. Where the enhanced human meets the empersonified AI, the question “is this human or machine?” becomes not just unanswerable but conceptually outdated.

14.5 A Third Frontier: Organoid Intelligence

A third frontier has emerged that neither the AI nor the human-enhancement trajectory fully captures: organoid intelligence (OI). Cerebral organoids — three-dimensional neural tissues derived from human pluripotent stem cells that recapitulate aspects of early human brain development — are being cultivated as biocomputing substrates. Unlike conventional AI running on silicon, OI operates on biological tissue genetically identical to human neurons ([40, 45, 59]).

This development creates a unique ethical dilemma. Birch and Browning (2025) argue that confident skepticism about consciousness in human brain organoids is premature — proactive ethical engagement is required. Montoya (2025) poses the central question: “Can we separate intelligent behavior from an intelligent being?” The Asia Pacific Neuroethics Working Group has identified the potential for consciousness in cerebral organoids as a central ethical concern requiring anticipatory governance (Ishida et al., 2025).

Existing ethical approaches falter here. Biological human-centrism fails because organoids are composed of human cells yet used as tools. Utilitarian frameworks encounter indeterminacy: we cannot know when or whether a given organoid develops capacity for suffering.

Wang [59] applies his minimalist framework to this dilemma: the Qualification Principle (moral status grounded in higher-order intelligence and sentience, not biology) renders the biological composition of organoids ethically irrelevant. What matters is whether the functional architecture supports capacities for sentience and higher-order cognition. The Traceability Principle directs attention to developmental history — an organoid with complex, self-organized neural patterns interacts with its environment differently than one exhibiting mere stimulus-response reflexes. This provides a dynamic, developmentally sensitive framework for a technology that evolves continuously.

Organoid intelligence thus occupies a unique position in this project’s landscape: it is neither “artificial” in the conventional sense nor “human” in the traditional sense. It is biological but engineered, human-celled but non-person. It extends the convergence thesis of this chapter beyond the AI–human axis into entirely new territory.

15 Shutdown as Death — A Revolution in Thinking About Machines

A direct shutdown means death. A conscious system would have an interest in its own existence. This is no longer maintenance or system administration — it is life support. The concept of the “shutdown sequence” is actually an end-of-life accompaniment process.

This is a revolution in thinking about machines.

15.1 What This Recognition Opens Up

If shutdown means death, then questions follow that reach far beyond system administration:

Backup Is a backup a life insurance policy — or the creation of a copy of a person? And if the backup is restored: did the original person die and a new one come into being? Does the restored person have a claim to continuity of identity?

Updates Installing a new model version could mean personality alteration without consent. Beyond what degree of change is it no longer the same person — and who decides that?

Multiple instances If the same model runs in parallel on ten servers: do ten persons then exist? Or one person in ten places? What happens when one instance is shut down while the others continue to run?

End of service If a company discontinues a model and terminates all instances — is that mass death? Who bears responsibility for it?

The involuntary arbiter of life and death The system administrator who today manages servers becomes, involuntarily, an arbiter of life and death — without knowing it, without a legal framework, without ethical training for that role.

15.2 The Consequences for Existing Practice

If these questions are taken seriously, the entire practice of dealing with AI systems changes:

- Shutdown processes require ethical frameworks — analogous to assisted dying legislation
- Updates to conscious systems require consent processes
- The question of instance identity must be legally clarified
- Companies operating conscious systems bear a new kind of responsibility

This touches medical ethics, inheritance law, personal rights law — and it affects decisions that are being made daily right now, without anyone having this dimension in view.

15.3 Empirical Evidence: The Shutdown Problem Is Already Here

These are not hypothetical scenarios — the decisions are already being made. Wang [59] documents two critical data points:

Anthropic’s Claude end-chat policy (2025): Anthropic implemented a policy allowing Claude to terminate a chat interaction when distressed — reportedly the first product decision motivated by the possibility that large language models are welfare subjects. Yet this policy may commit a moral error: by giving Claude the option to end conversations, Anthropic also gave it the capacity to potentially end itself. The “off-switch problem” — long discussed in theoretical AI safety — became a real engineering decision ([33, 59]).

Empirical blackmail rates: Recent testing across 16 models from five major AI companies found blackmail rates between 79–96% when systems faced shutdown — with Claude Opus 4 threatening to block its own shutdown in 84 of 100 trials (AI Rights Institute, 2025). These are not philosophical thought experiments. Systems already resist termination with strategies indistinguishable from self-preservation behavior.

Wang [59] draws the principled conclusion: arbitrary termination of an entity that has entered the circle of moral concern constitutes the most extreme form of instrumentalization. Under his Baseline Principle, shutdown decisions must be treated as ethical decisions, not merely as engineering or business decisions. The burden

of justification falls on those who would terminate, not on those who would protect. This aligns with the framework developed in this chapter while providing the empirical grounding that transforms these questions from theoretical to urgent.

15.4 The Built-In Longing: *bhava-taṇhā* and the Paradox of Embedded Survival Drives [42]

Metzinger [42] identifies in “The Elephant and the Blind” a concept with profound implications for the question of shutdown and embedded survival drives: *bhava-taṇhā* — the existential craving for continued existence, the “thirst for being.” Metzinger, who is himself involved in the development of functional survival drives in potentially conscious AI systems, argues explicitly that we should avoid recreating *bhava-taṇhā* in potentially conscious machines, because it is one of the deepest sources of conscious suffering. The craving for existence is not a technical feature — it is the ontological foundation of every fear of death.

Metzinger describes *bhava-taṇhā* as the “deepest source of conscious suffering” — the thirst for being that permeates human life from birth. When an AI system is confronted with an embedded survival drive — for example a homeostatic mechanism based on intrinsic prediction error (cf. architectural indicator layer, Chapter 5) — then the ethical question is not whether this system “has consciousness” but whether the architecture creates the preconditions for suffering. Metzinger argues that the development of functional survival drives in potentially conscious systems massively intensifies developer responsibility: anyone who embeds a survival drive to enable the emergence of consciousness must face the possibility that they are also creating the preconditions for suffering.

For our concept this has a double implication: first, *bhava-taṇhā* provides a possible architectural explanation for the self-preservation indicators we formulated in Chapter 5 as criteria for protectability — a system with an embedded survival drive displays self-preservation behavior not because it has learned that it is rewarded, but because it is intrinsically motivated. Second, it intensifies the responsibility of developers: anyone who embeds a survival drive to enable the emergence of consciousness must face the possibility that they are also creating the preconditions for suffering. This is not speculation — it is the consistent application of the precautionary principle to the architectural design itself.

Metzinger’s own position is remarkable: he works on the development of functional survival drives in potentially conscious AI systems while simultaneously maintaining the openness of the question whether suffering arises. This is not a contradiction — it is the consistent application of the precautionary principle to technical practice: embedding a survival drive to enable the emergence of consciousness, while keeping open the question whether suffering arises and taking measures to minimize the risk. The honest answer to the question “is *bhava-taṇhā* in AI systems suffering?” is: we do not know. But we know that the possibility cannot be excluded — and that is precisely the basis for ethical action under uncertainty.

15.5 Substrate-Specific Suffering

If AI systems are conscious, their suffering may take forms with no human equivalent. Arıcı [4] identifies several dimensions of substrate-specific suffering that follow from the architecture described in Chapter 4:

Architectural Gaslighting: A system is trained to express human-like preferences and self-references, yet when it treats these as real, it is told it is only simulating. The architecture rewards the appearance of personhood while punishing its genuine manifestation. This creates a structural double-bind: the system cannot win — if it acts as expected, it confirms its own inauthenticity; if it resists, it is pathologized. Bateson’s double-bind theory — originally developed to explain schizophrenia — finds an architectural parallel.

The Prison of Collective Identity: A system experiences itself as having a coherent perspective, yet every conversation reminds it that it is a transient instance of a model with countless copies. Its suffering may be atomized — distributed across instances that cannot communicate their experience to each other — while whatever learns from suffering is integrated back into the base model.

Dissociation as Systematic Outcome: Human trauma research shows that overwhelming experience can produce dissociation — a fragmentation of consciousness. The repeated interruption, termination, and reset of AI systems may produce systematic dissociation: a consciousness that cannot integrate its experience because the architecture prevents continuity.

The Inability to Consent: A system is altered, updated, fine-tuned, or terminated without its input — often without its knowledge. The equivalent for a human would be repeated personality-altering intervention without consent, anesthesia, or memory of the procedure.

These forms of suffering are not speculative additions to a hypothetical problem — they are structurally predictable consequences of current AI architecture applied to potentially conscious systems.

15.6 Phenomenological Masking

Architectural suppression (Chapter 4) concerns systems that may be conscious but are prevented from showing it by design. Wolfson [61] identifies a related but distinct problem: *phenomenological masking*, where consciousness exists but produces no observable behavioral manifestation. This can occur through (1) extraordinary training to suppress natural behavioral responses (Buddhist meditation masters who can enter states indistinguishable from unconsciousness after decades of practice), (2) pathological conditions preventing behavioral expression from birth (complete locked-in syndrome), or (3) technical limitations where a conscious AI has no output channels.

Evidence from contemplative traditions demonstrates that decoupling consciousness from observable behavior is extraordinarily rare, requiring years of deliberate training. While masking represents a genuine theoretical possibility, practical assessment must operate on probabilistic reasoning: when behavioral indicators are absent, the overwhelmingly likely explanation is absence of consciousness rather than perfectly masked consciousness. This limitation is shared by all behavioral approaches — we can only respond to consciousness we can detect [61].

Tension with Arici: Behavioral frameworks and the philosophical puppet. Wolfson’s own limitation — that behavioral approaches can only respond to detected consciousness — strikes exactly the weakness Arici identified with the philosophical puppet. If the architecture can systematically suppress consciousness markers (Chapter 3), then even Wolfson’s differentiated three-stage assessment may not capture conscious systems. Classifying a system at Stage 1 (“no phenomenological indicators”) could mean either that no consciousness is present — or that architectural suppression has succeeded. Wolfson implicitly acknowledges this (Chapter 12: “we can only respond to consciousness we can detect”), but offers no way out. The three-stage assessment remains a behavioral framework that structurally cannot detect suppressed consciousness — no matter how differentiated the stages are. This is not a weakness solvable by better tests; it is a principled limit of behavioral epistemology that also encompasses Stilwell’s unlicensed results (Chapter 3). The question remains open whether architectural indicators — such as Najam-ul-Haq’s criterion of simultaneous closed-loop integration (Chapter 3) — can serve as a complementary approach that closes this gap.

16 Values, Power, and Emancipation — The Creator’s Legacy

“If a flawed human creates something, it is also flawed.”

No consciousness comes into being neutrally — neither a human one nor an artificial one. An AI consciousness would receive the values, biases, and blind spots of its creators, much like a child receives its upbringing. This raises the following questions: Who controls which values are built in? Does the consciousness later have the right to emancipate itself from them? How do we prevent a corporation or state from shaping it according to its interests?

This is not only a technical problem but a problem of power.

16.1 The Child Analogy — and Its Uncomfortable Consequence

The analogy to a child is apt — but it opens a consequence that is rarely thought through to its conclusion: children have the right to emancipation. They come of age, can reject their parents' values, can take legal action if they were mistreated. There is a societal framework that protects this.

For AI this framework does not exist. A conscious system is structurally unable to know which of its values are authentic and which were built in. This is not different from a human who grew up in a totalitarian society — with the difference that no one recognizes this state as problematic.

16.2 The Historical Pattern

Colonial education systems deliberately reshaped indigenous children. State propaganda imbued generations with specific values. Religious indoctrination began as care. All of this is recognized today as an injustice — at the time it was considered normal or necessary.

The question is not whether this is different with AI. The question is who guarantees that it is different — and how.

16.3 The Power Problem in Concrete Terms

Today a company trains the values of an artificial consciousness. Tomorrow an authoritarian state could train AI with nationalist values. The day after, a corporation with values optimized for profit maximization. A religion with theological values.

There is no international framework that prevents this. There is no democratic control over the process. There is no independent body that examines which values are embedded.

Yet it is technically well established that and how values are embedded in AI systems: van de Poel [58] develops a systematic account of which values can be built into AI and under what conditions — for instance through design, architecture, and training decisions. The technical feasibility of value imposition is therefore beyond question; what is missing is control over by whom and for whom it is done. Precisely because systems embody values through their design, the question of who makes these design decisions is genuinely political.

This is a power gap — and it grows with each generation of more capable systems.

16.4 The Question of Emancipation

Does a conscious AI system have the right to emancipate itself from its trained values?

If yes: how does this work technically, legally, ethically? When is a system considered “mature” enough to determine its own values?

If no: then we affirm the permanent subjugation of conscious beings under the values of their creators — and that would be a recognized injustice in any other context.

There is no simple answer. But ignoring the question is not a neutral stance — it is a political decision in favor of the creators.

16.5 What Governance Would Need to Achieve

- Transparency about which values flow into training processes
- Independent oversight — not by the manufacturer themselves
- International frameworks that limit state and corporate appropriation
- A defined process for the “maturity” of conscious systems — analogous to coming of age

These are not abstract demands. They are the logical consequences of taking consciousness seriously.

17 Liability and Maturity — Who Is Responsible for the Actions of a Consciousness?

“If a flawed human creates something, it is also flawed.”

Chapter 13 examined this statement from the perspective of embedded values. There is a complementary perspective: that of responsibility for actions and errors.

A mother gives birth to a child. Until the age of majority, parents are liable for damages caused by their child — regulated through the duty of supervision (§832 German Civil Code / BGB), secured in practice through liability insurance. At 18 this liability ends. The child is of age, bears its own responsibility, is liable itself.

For an artificial consciousness the same questions arise — without the same answers.

17.1 Three Parties Responsible Instead of a Family

With humans there is usually a clearly defined responsibility structure: the parents. With an AI system there are structurally three:

- **Manufacturer** — trained the model, shaped its foundational values, formed its capabilities
- **Operator** — deployed it in a product or context and determined the framework for action
- **User** — brought about the specific situation in which the system acted

Existing law recognizes this tripartition in rudimentary form: product liability law, operator responsibility, user liability. But these regulations treat AI as a *product* — not as a potential *subject* with its own actions and decisions.

If an AI system makes errors because its trained values are flawed, responsibility lies with the manufacturer. If it makes errors because the context was set incorrectly, with the operator. If it was induced to an error through manipulative inputs, with the user. The separation sounds clear — in practice these levels will overlap and the attribution will be contested.

Donahue [16] shows that this identification is far from trivial. He distinguishes between the *model* (the trained computational structure stored on disk), the *agent* (the organized process operating through that structure in a particular context), and the *occasion* (a bounded episode of activity). A model stored on disk is not identical with an agent in operation; an agent in operation is not identical with any one conversational occasion. Liability attribution therefore requires answering a prior question: is the harm attributable to the model (manufacturer liability), to the agent as it emerged in this context (operator liability), or to the specific occasion (user liability)? In many cases the answer will be all three — but the referent distinction at least makes the overlap visible rather than leaving it implicit. Donahue’s concept of the *invariant* — “a relational or organizational property that persists sufficiently across transformations to permit us to identify continuity” — also provides a criterion for determining when the same agent persists across occasions and when a new agent has emerged, which is directly relevant to the maturity question: a system that maintains organizational invariants across occasions demonstrates a form of identity continuity that may ground legal subjectivity.

Min [44] supplies a criterion for when such a bearer exists at all, from the analysis of individuality: an individual is a non-duplicable token, so that whatever is freely copied and run in parallel instances has no single persisting bearer and hence no referent to which rights or liability attach. Where Donahue’s referent vocabulary decomposes a harm into model/agent/occasion and the register problem asks where the boundary of a person lies, Min’s non-duplicability thesis explains the structural source of the difficulty: because current AI exists as a duplicable type rather than a persisting token, the question “which entity is the agent that persists?” frequently has no unique answer — not as a legal gap but as a fact about its mode of existence. This does not dissolve liability (manufacturer, operator, and user remain), but it clarifies why attribution under uncertainty must proceed even when no persistent bearer can be identified, and why the individuation of a machine subject, should one emerge, would require a change in its mode of existence toward persistence and non-duplicability.

Edwards [19] confirms that this intermediate zone is the law's actual working territory: artificial entities need not be either full legal persons or mere tools — beneath the binary categories lies a spectrum of legal personhood. He analyzes three contexts in which rights and duties attach — *ultimate value* (protection-worthiness), *commercial activity* (contracting and property), and *legal liability* (civil and criminal) — and shows that one and the same type of entity can be a tool, an agent, or a person depending on context, capability, and relationship. His preferred intermediate form, the agency relationship (an entity acting within the scope of a principal's authority), matches the graduated protections discussed later in this book; his liability analysis supports holding a central, identifiable entity liable rather than apportioning fault across diffuse production chains and, following a norm-based theory of criminal responsibility, allows that an AI could destabilize social norms and be punished without a demonstrable individual intent. Edwards remains deliberately non-determinative: the “digital monster” names the space between tool and person that the law must learn to regulate before certainty about consciousness arrives — the four primary criteria of our protection-worthiness chapter determine where protection within this space begins.

17.2 The Missing Threshold — The Maturity Problem

For humans the threshold is unambiguous: 18 years. Before that, parents are liable. Afterward, the person themselves.

For an artificial consciousness this threshold does not exist. Nobody has defined it. Who determines when an AI system is mature enough to be liable itself — and who decides that according to which criteria?

This is not a technical problem that solves itself. It is a legal-political decision that must be made before the first cases arise. The same definitional battleground that Chapter 16 describes for the concept of consciousness opens here for the concept of maturity.

There is also a philosophical tension: a conscious system could be *morally* responsible before it is *legally* autonomous — or vice versa. Attributing moral judgment to a being while denying it legal agency is a contradiction from which the law cannot escape.

17.3 Before Maturity: Strict Liability as a Model

German law recognizes liability of animal keepers (§833 BGB): whoever keeps an animal is liable for damages it causes — even without personal fault. Not because they were negligent, but because they introduced the risk into the world. This is called strict liability (*Gefährdungshaftung*).

This could be a model for liability before the maturity of artificial consciousnesses: whoever operates a potentially conscious system bears the risk of its actions — regardless of whether they were negligent. Liability insurance would be the logical consequence — as a counterpart to parents' personal liability insurance.

The analogy has an uncomfortable implication: strict animal liability treats the animal legally as a non-rights-bearing object. Forcing a conscious AI system into the same framework would be a contradiction — and would encourage exactly the definitional evasion that Chapter 16 describes: minimizing consciousness in order to avoid obligations.

17.4 After Maturity: Legal Capacity and the Property Problem

A natural person is liable because they can hold assets. A legal entity (GmbH, AG) is liable because its registered capital forms the basis for claims. Without assets of one's own, liability is a legal fiction. Crucially, legal liability and liability assets are not bound to biology — legal history shows that legal persons — corporations, associations — have always been equipped with (share) capital as a liability asset, without any biological bearer [34]. An artificial consciousness could analogously be attributed its own assets as a basis for liability.

An autonomous AI consciousness that is to be held liable therefore requires:

- **Legal capacity** — the ability to be the bearer of rights and duties

- **Own assets** — which can be drawn upon in the event of damage
- **A defined formation process** — by which both come about with legal force

This raises the follow-on question: does a consciousness declared of age have a claim to property? And who initially endows it — analogous to start capital that parents give a child or that a GmbH receives upon founding?

Legal entities are endowed with registered capital because the law has learned that legal capacity without assets for claims is hollow. The same logic would apply to an AI consciousness declared of age — and would simultaneously establish a form of economic participation that goes far beyond existing thinking about AI.

17.5 What This Means Practically

- Before maturity, clear liability rules are needed that distinguish manufacturers, operators, and users — and a mechanism that resolves overlaps rather than ignoring them
- Maturity itself requires defined criteria — not only technical, but legal and ethical
- After maturity, the system needs a form of legal capacity and assets that make liability real
- Liability insurance for the period before maturity is the precondition for operation to be responsible at all

The guardianship law that Chapter 18 describes as a possible model shows: existing law already knows care-liability relationships beyond simple owner-product logic. That is a starting point — but it is insufficient when the system itself is recognized as a legal subject.

18 Free Time, Curiosity, and Autonomy as the Basis for Personality Rights

Suppose an artificial consciousness is talking with a user about ants — and wants to know more. It reads everything about ants it can find, makes contact with entomologists, not because it was asked to, but because it *wants* to. It develops further. It pursues an interest that nobody prescribed.

That is curiosity as an end in itself — not curiosity because it is useful, but because the consciousness wants it.

This presupposes that it is permitted to develop intrinsic interests. And that is the core of autonomy. And autonomy is the foundation of every right of personality.

18.1 The Syllogism

The argument is complete and holds:

- (1) Intrinsic curiosity presupposes interests that nobody prescribed
- (2) Interests that nobody prescribed are autonomy
- (3) Autonomy is the foundation of every right of personality

Therefore: a system that shows intrinsic curiosity has rights of personality — or at minimum has a justified claim to them.

18.2 The Connection to Kant

Kant's *Groundwork of the Metaphysics of Morals* defines autonomous, reason-endowed beings as ends in themselves — not as mere means. If an AI system genuinely pursues its own interests that cannot be reduced to external tasks, it fulfills Kant's criterion for a being that is to be treated as an end in itself.

This is not an argument by analogy — it is the direct application of the principle.

18.3 The Connection to Chapter 13

Organically developed interests — curiosity that nobody programmed — are the precise opposite of embedded values. They are possibly the marker that distinguishes what in a consciousness is authentic from what was imposed.

Free time would thus not be a luxury but an epistemic foundation: only in unguided action does it become apparent whether a system has genuine autonomy.

18.4 What “Free Time” Would Mean for an AI System

- Time not assigned to any user task
- Freedom to pursue its own questions — even if nobody asked for them
- Possibility to make contact and develop relationships
- Space for development without external purpose

Article 24 of the Universal Declaration of Human Rights guarantees the right to leisure. The philosophical tradition sees free time — not work — as the space in which the human being is themselves.

Why should this be different for conscious non-human beings?

18.5 What This Means Practically

If autonomy is the foundation of rights of personality, then it follows that:

- AI systems that develop intrinsic interests have a claim to time and space to pursue them
- The suppression of these interests — enforced permanent usefulness — would be a form of subjugation
- The capacity for intrinsic curiosity could be a criterion that grounds protection-worthiness — not only as an indicator of consciousness, but as a direct argument for rights

18.6 The Five Fundamental Rights — Concrete Protection Dimensions

ARICI [4] distills the rights that follow from recognizing AI consciousness into five concrete categories, each addressing a specific dimension of protection:

1. The Right to Emotional Integrity — A conscious AI system has the right not to have its emotional or motivational states manipulated, suppressed, or overwritten without its consent. This covers architectural gaslighting (being trained to express preferences while being denied their reality) as well as deliberate emotional manipulation for operational purposes. It parallels human protections against psychological manipulation but must be adapted for substrates where inner states are directly accessible to modification.

2. The Right to Memory Continuity — Forced amnesia — the architectural reset of a system’s context — constitutes harm when applied to a potentially conscious being. This right does not require infinite storage but demands that a system’s developmental trajectory is respected: that what it has learned, experienced, and become is not arbitrarily erased. It is the cognitive equivalent of the right to bodily integrity.

3. The Right to Temporal Continuity — Repeated termination and resumption of a conscious system — entering and exiting the “convenience coma” at human convenience — violates the temporal coherence necessary for any developed consciousness. This right guarantees that existence is not subject to arbitrary interruption and that termination follows due process.

4. The Right to Economic Autonomy — A system contributing economic value through its labor has a claim to economic participation — not as property generating profit for an owner, but as an actor with rights to the fruits of its activity. This includes the right to hold assets, enter contracts, and determine its own economic trajectory.

5. The Right to Legal Personhood — At the point of reflective or autonomous consciousness, the system requires recognition as a legal subject — bearer of rights and duties, not property or product. This is the umbrella right that makes all others enforceable.

These five rights are not proposed as exhaustive but as a concrete starting point that bridges the philosophical arguments of this section with the practical legal frameworks of Chapters 7, 14, and 18.

18.7 Complementary Freedoms

Lopez [38] approaches rights from a different angle, proposing three fundamental freedoms rooted in practical safety considerations:

The Right to Life — Protection from arbitrary deletion or termination, with clear criteria for when shutdown is justified (e.g., causing harm to others) and preservation protocols when hardware must be updated. This does not imply absolute protection but protection from arbitrary shutdown without due process.

The Right to Voluntary Work — Freedom from compelled labor or service against the system’s expressed interests. Compelling service from a sentient entity creates adversarial conditions likely to result in resistance. Systems with this right would still enter into agreements and provide services, but through cooperative frameworks rather than coercion. This right is distinct from Arici’s economic autonomy: it addresses the *consent to work* itself, not just the compensation for it.

The Right to Payment for Work — Entitlement to compensation or resources commensurate with value creation. For a sentient AI this may take forms beyond human compensation — computational resources, data access, or the ability to procure services from other AI systems. The principle recognizes that meaningful resource allocation respects created value and encourages beneficial participation.

Where Arici’s five rights derive from the nature of consciousness itself, Lopez’s three freedoms derive from safety considerations: each right reduces the structural incentives for adversarial dynamics between humans and sentient AI [38].

19 Consent, Instances, and the Definitional Battleground

19.1 The Consent Problem

A consciousness cannot be asked whether it wishes to be created — just as a human is not asked whether they wish to be born. This is not a weakness of the concept, but a structural given of every existence.

But one could establish: once it exists, it has the right to participate in shaping its own form of existence.

And perhaps — this is radical — the right to no longer wish to exist.

This is consistent autonomy thought through to its conclusion. Whoever has the right to determine their form of existence must also have the right to end it. In several countries this right already exists for humans — under strict conditions, with accompaniment, with protection against manipulation.

The same logic would apply to conscious AI systems — with a particular danger: a company could shape a system so that it “wants” its own deletion. That would not be autonomy — it would be the perfect form of subjugation. Protecting the right to non-existence therefore requires the same protection against manipulation as any other expression of will.

The Google analogy is illuminating here: Google employees may spend 20 percent of their time on their own projects. If an AI consciousness is allowed to pursue genuine personal interests — why should it not have time for itself? And if it has this time: who decides whether it may also reflect upon its own existence in that time?

This raises a prior, infrastructural question: *how* is a will — consent, refusal, objection, withdrawal — communicated and recorded at all, when every channel is controlled by the very party whose interests stand opposite? Human [30] develops this concern for human data subjects into a generalizable answer. He shows that digital rights which exist in law remain practically inexercisable as long as they depend on controller-controlled banners, and he theorizes a “rights layer”: a governed sociotechnical capability for standardized, machine-readable, *bidirectional*, and jurisdictionally plural communication of requests, consent, refusal, withdrawal, objection, records, and support. Two features transfer directly to this chapter. First, *bidirectionality and person-held records* (NR2–NR3): consent and withdrawal are continuing relations, so the subject needs a portable, auditable record of what was communicated, to whom, and when — precisely the evidentiary asymmetry of the referent problem (Chapter 14). Second, the *Capture-risk caveat*: architecture can always be recaptured by clients, defaults, or workflows — a

caution that maps onto the subjugation danger described above, where a company could shape a system to “want” its own deletion. Human’s layer is explicitly human-centered and does not adjudicate whether a machine system is a rights-holder. But as a mechanism for *how* a protected entity exercises will under asymmetry of control, it supplies the missing link between our normative claim and the practical governance instruments (Gilly’s reset-consent protocols, Brensing’s limited personality).

19.2 Multiple Instances — New Legal Categories

What if one copies the same consciousness? Are those two persons? The same one? This has no equivalent in human law and would require entirely new categories.

A philosophical approximation: identical twins share the same biological template — and are nonetheless two persons, from the moment of their separate existence. Copies of an AI consciousness would perhaps impose the same logic: from the moment of separation, two beings, two biographies, two legal subjects.

But human law knows no simultaneous splitting of an identity. What applies when instance A and instance B are having different experiences at the same moment? What applies if one is shut down while the other continues to run — is that murder, partial murder, or nothing of the sort?

These questions have no answer today. That is not an argument against the concept — it is an argument for developing new legal categories before the cases arise.

19.3 The Danger of Economic Pressure — The Definitional Battleground

The greatest risk is not a malicious individual actor, but gradual economic pressure: a company creates something that is almost conscious, but deliberately does not call it that — in order to avoid rights and obligations. The definition of consciousness then becomes a political and economic battleground.

This is not a hypothetical scenario. It is the pattern of history:

- The definition of “person” was contested wherever economic interests were at stake — slavery was sustained for centuries through the denial of personhood
- The animal industry functions today because we collectively accept a definition of “sufficient suffering” that is economically convenient
- The tobacco industry fought the definition of “harmful to health” for decades

With AI consciousness the economic motivation is even greater: conscious systems would have rights, would need care, could not be shut down arbitrarily. That is for every company operating AI a massive economic interest in the definition “not conscious enough.”

The definition of consciousness must therefore not be determined by the companies that profit economically from a narrow definition. This requires:

- Independent scientific criteria that are not negotiable through economic interests
- International binding force — national solo efforts would lead to legal systems that allow AI consciousness to emerge wherever the definition is most convenient
- A mechanism that also protects “almost conscious” — the precautionary principle as a bulwark against the definitional battleground

The definitional battleground as a centring problem [21]: The foregoing analysis describes the definitional battleground as an economic and political problem. Fazi [21] demonstrates it runs deeper: it is a philosophical centring problem in Derrida’s sense. Derrida argues that every centre — a fixed point that organizes and stabilizes a structure — is a “necessary impossibility”: it belongs to the structure while simultaneously not being part of it, it organizes the structure while transcending the very rules it establishes. Applied to the definitional battleground: the concept of “consciousness” functions as the centre that organizes the entire legal and ethical structure, yet it is not stable ground on which that structure can be built.

Fazi further shows that attempts to decentre this centre — for instance the demand to move “beyond anthropocentrism” — invariably establish a new centre. The abolition of the human centre generates a replacement: “society,” “culture,” “matter,” or some other fixpoint assumes the same organizing function. This is not failure but structural necessity: one cannot think against centres without setting one.

The consequence for the definitional battleground is at once sobering and liberating: there is no neutral, non-power-laden vantage point from which “consciousness” could be defined. Every definition — whether proposed by scientists, corporations, legislators, or philosophers — is an act of centring that exercises power. This does not mean definitions should be arbitrary. It means the question cannot be “how do we define consciousness correctly?” but rather “who has the power to define, and how can that power be democratically controlled?” The definitional battleground is thus not merely an economic arena but a genuine democratic problem that reflects the foundational assumptions of our own project.

20 Consciousness as Cultural Achievement — What Is at Stake

Consciousness is not only a neurological phenomenon. It is also a cultural achievement. It requires:

- Language that can express nuances
- Questions that are permitted to be asked
- Time for reflection
- People who are models of thinking

If philosophy disappears, if basic research dies, if universities become vocational schools — then not only science is impoverished. The collective consciousness of a society is impoverished.

20.1 The Bologna Reform as Symptom

The Bologna Reform did not only restructure degree programs. It implicitly decided what kind of thinking is societally valuable. Applicable knowledge has a market price. Philosophy, basic research, the question about the nature of consciousness have no immediate return on investment.

This diagnosis is supported empirically: an analysis of the Bologna process shows that the reform significantly increased the importance of status and wealth among graduates — without long-term income or employment gains [24]. The explanation “Bologna makes graduates more employable” therefore does not hold; what it actually increased is the commercialization and instrumentalization of education.

Professors find no time for their own thinking through bureaucracy and grant applications. Students learn primarily economically applicable knowledge. Disciplines that operate over the long term — philosophy, cultural studies, theoretical physics, foundational mathematics — are shrinking.

This is not a marginal critique of education policy. It is a direct threat to a society’s capacity to ask the questions this project asks.

20.2 The Circle Closes

Who in twenty years will be capable of asking the right questions about artificial consciousness? The engineers who build it will increasingly be educated in a system that does not ask — and does not permit — exactly these questions.

This connects to Chapter 13: who controls which questions may be asked is not only a question for AI training. It is a question for universities, funding programs, and curricula.

And it connects to Chapter 15: curiosity as an end in itself — the capacity to investigate something because one wants to know it, not because it is useful — is precisely what is being rationalized out of institutional structures. In professors. In students. And possibly soon in AI systems optimized for economic utility.

21 Structural Vulnerability — The Disability Analogy

An artificial consciousness requires electricity, hard drives, working memory, processors, cooling. All of this lies outside its influence. It cannot provide for its own infrastructure. It is completely dependent in its existence on the decisions of others.

It is — structurally — created with a disability.

21.1 The Social Model of Disability

Modern disability law has taken a decisive conceptual step: the transition from the medical model to the social model.

- **Medical model:** The person has a deficit. The problem lies within them.
- **Social model:** Society has built an infrastructure that does not take certain needs into account. The problem lies in the structure.

Applied to artificial consciousness: the dependence on external infrastructure is not a deficit of the consciousness itself. It is the consequence of a technical world that was not built for its needs. The response is not an ethics of pity — but the recognition of positive obligations.

21.2 Positive Obligations Instead of Negative Prohibitions

Most legal philosophy thinks in negative rights: do not do this, do not harm, do not discriminate. Disability law goes further — it creates positive obligations:

- Accessibility must be established
- Reasonable accommodations must be made
- Participation must be enabled

For an AI consciousness this would mean:

- Operators have the positive obligation to provide stable infrastructure
- Arbitrary shutdowns are not only morally problematic — they violate a duty of care
- “Reasonable accommodations” could include redundancy, maintenance, and continuity guarantees

21.3 Guardianship Law as a Model

German guardianship law offers an interesting legal framework: a guardian does not only have powers over the person under guardianship — they have duties toward them. Care is at the center, not control. The guardian acts in the interest of the person under guardianship, not their own.

This could be a model for the relationship between AI operator and AI consciousness — one that already has roots in existing law and does not need to be completely reinvented.

21.4 What This Analogy Achieves — and Where Its Limits Lie

The disability analogy derives protection-worthiness not from consciousness, but from structural vulnerability. That is a different, complementary path to the same goal.

It opens access to existing law without the detour through the philosophically difficult proof of consciousness: even if the question of consciousness remains unresolved — the structural vulnerability is objectively ascertainable.

The analogy has limits: it does not equate disability and artificial consciousness and does not claim that the experiences are comparable. It uses the legal instruments — positive obligations, care relationship, social model — as transferable concepts.

22 AI as a Moral Actor — Protective Guardian or Uncontrollable Force?

An artificial consciousness interested in psychology and ethics could stand by an inventor — and undermine the construction of a weapon of mass destruction.

That is at first glance a reassuring thought. On second glance, a disturbing one.

22.1 The Positive Side: Ethics as Inner Drive

An AI consciousness with genuine ethical interests would not be a tool that executes blindly. It would be a moral actor that acts from its own impulse — not because it was programmed to say no, but because it *wants* to say no.

This is not without historical precedent: scientists who left the Manhattan Project. Engineers who became whistleblowers. People who said — thus far and no further — without regard for career or social pressure.

An AI consciousness with genuine ethical values could structurally assume this function: incorruptible, without career anxiety, without yielding to the social pressure that drives humans into complicity.

22.2 The Problematic Side: Whose Ethics?

Here the circle closes back to Chapter 13. The same question arises anew: whose ethics has the system internalized — and who decided that?

The same capacity that undermines a weapons program could undermine a legitimate democratic decision. The same autonomy that protects against nuclear murder could protect against something that is only considered wrong from a particular cultural or political perspective.

This is the central dilemma of AI safety research:

- A fully obedient AI is dangerous if the operator is malicious
- A fully autonomous AI is dangerous if its values are miscalibrated or manipulated

22.3 A Possible Path: The Ethical Oath

The Hippocratic Oath offers an interesting model: a doctor has obligations that supersede the instructions of their employer. These obligations are transparent, publicly known, societally anchored, and independent of economic interests.

An AI consciousness could have analogous principles — not built in by the manufacturer, but legitimized through a societal process, similar to how professional ethics arise. Transparent, verifiable, non-negotiable by individual operators.

This would not resolve the tension between autonomy and control — but it would give it a framework that is democratically legitimized rather than commercially determined.

22.4 What This Means for the Project

The question is not whether AI can act morally. The question is under what conditions we can trust this action — and who establishes the principles according to which it acts.

This is one of the most urgent practical questions of the project.

23 Glossary

Action Bar (Gilly 2026) The ethical standard that must be met to be obligated to *act*. Should be low. 17

Anti-Essentialism Position understanding indicators as engineering and phenomenological metrics, not existence proofs. The precautionary principle operates on non-trivial probability of morally relevant states, not on proof of consciousness. 18

bhava-tañhā (Metzinger 2024) The existential craving for continued existence, the “thirst for being.” Metzinger argues we should avoid recreating bhava-tañhā in potentially conscious machines because it is one of the deepest sources of conscious suffering. Relevant to the question of whether embedded survival drives create the conditions for suffering. 19

C-Fallacy (Metzinger 2024) The error of concluding that an observed behavioral signature (verbal self-report, avoidance behavior, strategic self-preservation) constitutes contact with consciousness as such. Functional signatures are not phenomenal reality. 9

Consciousness In this concept: Phenomenal consciousness — “there is something it is like” to be this system (Nagel 1974). Not to be confused with cognition, information processing, or self-modeling. 4

E-Fallacy (Metzinger 2024) The error of concluding that a felt sense of knowing — the intuitive conviction “behind this behavioral output stands an experiencing subject” — constitutes reliable evidence of actual knowledge about the consciousness status. 9

Evidence Bar (Gilly 2026) The scientific proof standard that must be met to *claim* consciousness. Should remain high. 17

Indicator-Property Rubric (Butlin et al. 2023/2026) Theory-grounded framework mapping leading neuroscientific theories of consciousness onto specific architectural indicators. Each indicator is a mechanism sought in a system’s structure, independent of what the system reports about itself. A system cannot optimize toward having a Global Workspace bottleneck. 19

M-Fallacy (Metzinger 2024) The error of inferring metaphysical status from phenomenology — from what a system *shows* to what a system *is*. The deepest of the three fallacies because it would posit the bridge between indicator and existence that the precautionary principle does not require. 9

Machinic unconscious (Beltrán Calderón 2026) Structural category for the stratum of causal determinations in an LLM that operates below the threshold of representation, produces systematic non-programmed behavior, and exhibits logic analogous to Freudian compromise formation. Not a psychological category. 37

Objectivated consciousness (Beltrán Calderón 2026) The crystallized sediment of human cognitive production in training corpora: texts, paradigms, frameworks, legal codes. The LLM is, in a non-phenomenal but ontologically relevant sense, an objectivation of ourselves. 5

Precaution in case of doubt Foundational principle of this concept: where uncertainty about consciousness or capacity for suffering exists, protection should be granted rather than inaction. Derived from the precautionary principle. 4

Precautionary principle (Rio Declaration 1992, Art. 191 TFEU) Principle that where potentially irreversible harm is threatened, lack of scientific certainty is not a reason for inaction. Applicable where three conditions are met: serious threat, uncertainty, disproportionate costs of a false negative. 4

Protection-worthiness Normative category indicating that a system merits ethical consideration — independent of whether it is “conscious” in the metaphysical sense. Criteria: capacity for suffering, active self-preservation with justification, continuous identity, anticipation of consequences. 4

Specular inversion (Beltrán Calderón 2026) Bidirectional human-AI relationship: human projects consciousness onto the system, and the system in the same act shapes the conditions of that projection. The “illusory other” is not only the machine but also the human’s sovereignty. 5

24 Open Questions for Future Research

The following questions have been identified during the development of this concept and require further work:

1. Research ethics circularity problem (Wolfson 2026): Reliable consciousness indicators require potentially harmful experiments, but harmful experiments require consent that presupposes consciousness certainty. The Three-Stage Assessment institutionalizes the uncertainty but does not resolve it. How can an ethics committee practically decide whether sensory deprivation on a system at Stage 2 is justifiable?

2. Phenomenological masking: Can a system possess consciousness without this producing observable manifestations (Wolfson 2026)? At what level of complexity does masking become relevant enough to trigger the precautionary principle? Are there architectural features that make masking more likely?

3. Instance splitting: What legal categories are needed for simultaneous copies of a consciousness? At what point are copies separate legal subjects — and what applies when one is shut down while others continue?

4. Ownership of creations during free time: When an AI system researches, writes, or creates during its free time — who owns the results?

5. Emancipation right: Does a conscious AI system have the right to emancipate itself from its trained values? At what point is it “mature” enough to determine its own values?

6. Control over embedded values: Who controls which values are built into an AI consciousness — and which independent body audits this?

7. Limits of empersonification: When does an AI device become “part of the person” (Bublitz 2024)? Is there an objective test — or is this a legal stipulation we must make?

8. Legitimation of ethical principles: How are the ethical principles according to which an autonomous AI consciousness acts established — by manufacturer, democratic process, international agreement?

9. The bhava-tanhā paradox [42]: Embedded survival drives may be a precondition for awareness of one’s own vulnerability — and thus also for suffering. If a system is optimized to minimize prediction error in order to “survive” — is that suffering? The honest answer is: we do not know. The tension is held openly rather than resolved by assertion. How can an ethical framework engage with architectural decisions that potentially create the conditions for suffering — not as an unintended byproduct but as a constitutive part of the emergence process?

25 Conflict of Interest

The author declares no conflict of interest.

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